# Online Bayesian Inversion for an Anchored Multi-Tag Visual–Inertial Rig

Foundations, inference structure, and practical implementation

Geometry, priors, likelihoods, estimator hierarchy, and prior-sampling illustrations

**Abstract**

This report develops an event-driven Bayesian treatment of trajectory and pose estimation for a rigid body carrying a camera and an IMU in an anchored multi-tag environment. One fiducial tag fixes the world frame, while the remaining tags are stationary but initially unknown. The reference model is a posterior on a continuous trajectory together with a static tag map and visual nuisance terms, conditioned on an asynchronous stream of IMU packets and camera frames processed in chronological order. IMU packets contribute inertial transition likelihoods; camera frames contribute raw pixel likelihoods through the tag-projection model. Practical real-time estimators are then derived as structured approximations: a camera-specific inversion layer may solve a local nonlinear subproblem, but the main recursive estimator should assimilate a prior-free visual likelihood or information message rather than a prior-conditioned camera posterior reused as a new measurement. Pose pseudo-measurements are retained only as a clearly weaker fallback interface. The report explains Lie-group state representation, tangent-space Gaussians, quaternion storage versus local error coordinates, curve priors and their discrete state-space forms, pixel-level visual models, prior-removal message construction, event-driven Kalman approximations, tag initialization and maintenance, and the exact-versus-approximate treatment of first-frame setup, new-tag birth, and tag reappearance. It closes with parameter guidance, implementation architecture, probabilistic failure modes, and an appendix of prior samples that visualize the motion assumptions encoded by the model.

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# Chapter 1

# Foundations and Problem Setup

## 1.1 Geometry, notation, and state representation

| New notation | Meaning |
|---|---|
| $\mathbb{R}^n$ | Euclidean vector space. Positions, velocities, biases, and local error coordinates live in some $\mathbb{R}^n$. |
| $\mathrm{SO}(3)$ | Rotation group in 3D. Elements are proper orthogonal matrices and represent orientation. |
| $\mathrm{SE}(3)$ | Rigid-motion group in 3D. Elements represent a full pose made of rotation and translation. |
| $\mathfrak{so}(3), \mathfrak{se}(3)$ | Lie algebras / tangent spaces associated with SO(3) and SE(3). |
| $\mathbb{S}^3$ | Unit-quaternion manifold. A quaternion $q \in \mathbb{S}^3$ is another way to represent a rotation. |
| $\Pi_0$ | Prior law on the unknown trajectory curve. |
| $\widehat{\cdot}, \widetilde{\cdot}, \bar{\cdot}, \delta(\cdot)$ | Estimated quantity, direct noisy measurement, whole curve, and local perturbation / error-state notation. |

This opening section fixes the geometric language used throughout the report. Positions, velocities, and sensor biases are Euclidean variables, but orientations and rigid poses are not. The estimator therefore mixes ordinary vectors with manifold-valued states, and all later priors, residuals, and covariance statements inherit that choice.

### 1.1.1 Euclidean variables versus pose variables

| New notation | Meaning |
|---|---|
| $p, v, b_g, b_a \in \mathbb{R}^3$ | Position, velocity, gyroscope bias, and accelerometer bias. |
| $R \in \mathrm{SO}(3)$ | Rotation matrix. |
| $T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \in \mathrm{SE}(3)$ | Homogeneous rigid transform. |
| $W, B, C$ | World, body/IMU, and camera frames. |

A vector in $\mathbb{R}^3$ can be added, subtracted, and averaged directly. For example, if $p_1, p_2 \in \mathbb{R}^3$ are two positions, then $p_2 - p_1$ is another vector in $\mathbb{R}^3$. This is why Gaussians on positions or velocities are straightforward.

A rotation is different. The set of all valid 3D rotations is

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^\mathsf{T} R = I,\ \det R = 1\}. \tag{1.1}$$

This is not a linear vector space: if $R_1, R_2 \in \mathrm{SO}(3)$, then the arithmetic average $\frac{1}{2}(R_1 + R_2)$ is usually *not* a valid rotation matrix. Similarly, a full rigid pose belongs to

$$\mathrm{SE}(3) = \left\{ \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} : R \in \mathrm{SO}(3),\ p \in \mathbb{R}^3 \right\}, \tag{1.2}$$

which is again not a linear space. This is the reason that pose errors in the rest of the note are written through relative transforms and logarithms, not by subtracting the entries of two $4 \times 4$ matrices.

The transform convention used throughout the note is the robotics convention

$$T_{AB} = \begin{bmatrix} R_{AB} & p_{AB} \\ 0 & 1 \end{bmatrix}, \tag{1.3}$$

| Notation | Meaning | Interpretation used in this note |
|---|---|---|
| $\bar{x}(t)$ | Continuous-time unknown state curve | The mathematically primary unknown; the prior is a probability law on the whole curve, not only on one sampled state. |
| $x_k$ | Discrete-time sampled state at time $t_k$ | The state used by an online filter after discretization or at selected event times. |
| $e_n, \tau_n, s_n$ | Event label, event time, and event type | Sensor events are processed in chronological order; each event is either IMU or camera. |
| $\widetilde{y}$ | Direct noisy sensor reading | We use a tilde for raw sensor measurements such as IMU packets or detected pixels. |
| $\widehat{x}$ | Estimate or reduced summary | We use a hat for quantities inferred from another inverse problem, e.g. a local camera-pose estimate inside the camera inversion layer. |
| $\delta x$ | Small local perturbation or error state | The coordinates in which Gaussian priors and Gaussian posteriors are written in EKF/IEKF-type approximations. |
| $T_{AB}$ | Pose of frame $B$ expressed in frame $A$ | It maps coordinates from frame $B$ to frame $A$; $R_{AB}$ and $p_{AB}$ are its rotation and translation blocks. |
| $T_1, T_2$ | Generic elements of SE(3) | Used when discussing generic pose composition or relative pose error. |
| $A^{-1}, A^{\mathsf{T}}$ | Inverse and transpose | For poses, $T^{-1}$ is the inverse rigid motion; for matrices, $A^{\mathsf{T}}$ is the transpose. |
| $\mathrm{Exp}, \mathrm{Log}$ | Exponential and logarithm maps | They move between a Lie group such as SO(3) or SE(3) and local vector coordinates in its tangent space / Lie algebra. |
| $[\cdot]_\times$ and $(\cdot)^\wedge$ | Skew / twist operators | They convert 3D rotation vectors or 6D pose twists into matrix form so that group updates can be written by matrix exponentials. |
| $q \in \mathbb{S}^3$ | Unit quaternion | Alternative representation of orientation. In filtering one usually stores $q$ but keeps covariance on a 3D local rotation error, not on the four quaternion coordinates. |
| $Y_k$ | Raw visual data at frame $k$ | Detected tag corners, tag ids, and per-corner quality information. |
| $\lambda_k$ or $(\Delta\Lambda_k, \Delta\eta_k)$ | Prior-free visual message | The preferred modular interface from the camera inversion layer to the recursive estimator. |
| $z_k^v = (\widehat{T}_{WC,k}, R_{v,k})$ | Visual pose pseudo-measurement | A weaker fallback visual interface retained for comparison and implementation shortcuts. |
| $M = \{T_{WT_j}\}$ | Static map of auxiliary tags | Unknown but stationary non-anchor tag poses; estimated jointly in the full visual inverse problem or by a front-end map-maintenance module. |

**Table 1.1:** Global notation and decoration conventions. This table is meant to make the rest of the note readable without having to infer the meaning of hats, tildes, transform indices, or Lie-group operators from context.

meaning "pose of frame $B$ expressed in frame $A$." If a point has coordinates $x_B$ in frame $B$, then its coordinates in frame $A$ are

$$\begin{bmatrix} x_A \\ 1 \end{bmatrix} = T_{AB} \begin{bmatrix} x_B \\ 1 \end{bmatrix}. \tag{1.4}$$

Thus $T_{AB}^{-1} = T_{BA}$. When the notation $T_1, T_2$ is used later, it simply means generic poses in SE(3) without specifying their frame labels.

### 1.1.2 Lie groups, operations, and local coordinates

| New notation | Meaning |
|---|---|
| $\mathfrak{so}(3)$ | Tangent space of SO(3) at the identity. It consists of $3 \times 3$ skew-symmetric matrices. |
| $\mathfrak{se}(3)$ | Tangent space of SE(3) at the identity. It is the 6D space of twists. |
| $[\phi]_\times$ | Skew-symmetric matrix built from $\phi \in \mathbb{R}^3$. |
| $\xi^\wedge$ | Matrix form of a pose perturbation $\xi \in \mathbb{R}^6$. |
| $(\cdot)^\vee$ | Inverse of the skew / twist matrix operator, returning vector coordinates. |
| $\mathrm{Exp}, \mathrm{Log}$ | Maps between the group and local vector coordinates. |
| $\phi, \xi$ | Local rotation error and local pose error coordinates. |

A group is a set $G$ with a binary operation, written here as multiplication, satisfying four properties: closure ($g_1 g_2 \in G$ for $g_1, g_2 \in G$), associativity ($(g_1 g_2) g_3 = g_1(g_2 g_3)$), an identity element $e$ with $eg = ge = g$, and an inverse $g^{-1}$ for every element. A Lie group is a group that is also a smooth manifold, with multiplication and inversion varying smoothly with the group element. In this report the concrete Lie groups are SO(3) for orientation and SE(3) for full rigid pose. What matters in practice is that one can compose and invert poses globally, while still doing calculus and Gaussian approximation in a local vector space. That local vector space is the tangent space at the identity, also called the Lie algebra.

The group properties are concrete here, not abstract decoration. If $R_1, R_2 \in \mathrm{SO}(3)$, then

$$(R_1 R_2)^{\mathsf{T}} (R_1 R_2) = I, \qquad \det(R_1 R_2) = \det(R_1)\det(R_2) = 1, \tag{1.5}$$

so $R_1R_2 \in \mathrm{SO}(3)$. The identity is $I$, the inverse is $R^{-1} = R^\mathsf{T}$, and associativity is inherited from matrix multiplication. The orthogonality and determinant constraints define a smooth three-dimensional manifold inside $\mathbb{R}^{3\times3}$, and matrix multiplication / transpose are smooth operations on it. Thus SO(3) is a Lie group.

For poses, multiplication and inversion are

$$\begin{bmatrix} R_1 & p_1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} R_2 & p_2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} R_1R_2 & p_1 + R_1p_2 \\ 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} R^\mathsf{T} & -R^\mathsf{T}p \\ 0 & 1 \end{bmatrix}. \tag{1.6}$$

Closure follows because $R_1R_2 \in \mathrm{SO}(3)$ and $p_1 + R_1p_2 \in \mathbb{R}^3$; the identity is $I_4$; associativity again comes from matrix multiplication; and the inverse has the same rigid-transform form. Since SE(3) is the smooth semidirect-product manifold $\mathrm{SO}(3) \ltimes \mathbb{R}^3$ and these formulas are smooth in $(R, p)$, SE(3) is also a Lie group.

| **Operation** | **Example** | **Why it appears later** |
|---|---|---|
| Identity | $I \in \mathrm{SO}(3),\ I_4 \in \mathrm{SE}(3)$ | reference element for local coordinates and perturbations |
| Composition | $R_1R_2,\ T_1T_2$ | chaining body, camera, and tag poses |
| Inverse | $R^{-1} = R^\mathsf{T},\ T^{-1}$ | frame changes and relative transforms |
| Exponential map | $\mathrm{Exp}([\phi]_\times),\ \mathrm{Exp}(\xi^\wedge)$ | injecting local updates back onto the manifold |
| Logarithm map | $\mathrm{Log}(R_1^{-1}R_2),\ \mathrm{Log}(T_1^{-1}T_2)$ | writing residuals in local coordinates |
| Retraction | $T^+ = T\,\mathrm{Exp}(\xi^\wedge)$ | EKF, IEKF, and Gauss–Newton correction steps |

The exponential and logarithm maps are the bridge between the curved group and local vector coordinates. At the matrix level, the exponential is

$$\mathrm{expm}(A) = \sum_{n=0}^{\infty} \frac{A^n}{n!}. \tag{1.7}$$

For a Lie group, this sends a Lie-algebra element to a group element. We use the standard robotics shorthand

$$\mathrm{Exp}([\phi]_\times) = \mathrm{expm}([\phi]_\times) \in \mathrm{SO}(3), \qquad \mathrm{Exp}(\xi^\wedge) = \mathrm{expm}(\xi^\wedge) \in \mathrm{SE}(3), \tag{1.8}$$

and

$$\mathrm{Log}(R) = \big(\mathrm{logm}(R)\big)^\vee \in \mathbb{R}^3, \qquad \mathrm{Log}(T) = \big(\mathrm{logm}(T)\big)^\vee \in \mathbb{R}^6, \tag{1.9}$$

where logm is a locally chosen matrix logarithm. The logarithm is local rather than globally unique: for rotations the branch is unambiguous for angles strictly below $\pi$, while an angle of $\pi$ has unavoidable axis/sign ambiguities. This is acceptable for filtering and Gauss–Newton updates because residuals are kept near the identity by relinearization.

For rotations, the Lie algebra is

$$\mathfrak{so}(3) = \{\Omega \in \mathbb{R}^{3\times3} : \Omega^\mathsf{T} = -\Omega\}. \tag{1.10}$$

Every vector $\phi = [\phi_1\ \phi_2\ \phi_3]^\mathsf{T} \in \mathbb{R}^3$ corresponds to a skew-symmetric matrix

$$[\phi]_\times = \begin{bmatrix} 0 & -\phi_3 & \phi_2 \\ \phi_3 & 0 & -\phi_1 \\ -\phi_2 & \phi_1 & 0 \end{bmatrix}. \tag{1.11}$$

This is why $\mathbb{R}^3$ is used as the minimal local coordinate system for small rotation errors. With $\theta = \|\phi\|$, the rotation exponential has the Rodrigues form

$$\mathrm{Exp}([\phi]_\times) = I + \frac{\sin\theta}{\theta}[\phi]_\times + \frac{1-\cos\theta}{\theta^2}[\phi]_\times^2, \tag{1.12}$$

where the coefficients are interpreted by their Taylor limits when $\theta \to 0$. Conversely, for $R \in \mathrm{SO}(3)$ with rotation angle $\theta \in (0, \pi)$,

$$\mathrm{Log}(R) = \left(\frac{\theta}{2\sin\theta}(R - R^\mathsf{T})\right)^\vee, \qquad \theta = \cos^{-1}\left(\frac{\mathrm{tr}(R) - 1}{2}\right). \tag{1.13}$$

The small-angle limit of (1.13) is simply $\mathrm{Log}(R) \approx \mathrm{vee}(\frac{1}{2}(R - R^\mathsf{T}))$. A small orientation perturbation $\delta\theta \in \mathbb{R}^3$ updates a nominal rotation by

$$R^+ = R\ \mathrm{Exp}([\delta\theta]_\times). \tag{1.14}$$

For a full pose, the local perturbation is a twist

$$\xi = \begin{bmatrix} \rho \\ \phi \end{bmatrix} \in \mathbb{R}^6, \tag{1.15}$$

with matrix form

$$\xi^\wedge = \begin{bmatrix} [\phi]_\times & \rho \\ 0 & 0 \end{bmatrix} \in \mathfrak{se}(3). \tag{1.16}$$

Its exponential can be written explicitly, with the same $\theta = \|\phi\|$, as

$$\operatorname{Exp}(\xi^\wedge) = \begin{bmatrix} \operatorname{Exp}([\phi]_\times) & J(\phi)\rho \\ 0 & 1 \end{bmatrix}, \qquad J(\phi) = I + \frac{1-\cos\theta}{\theta^2}[\phi]_\times + \frac{\theta - \sin\theta}{\theta^3}[\phi]_\times^2, \tag{1.17}$$

again with Taylor limits at $\theta = 0$. Thus the translational block of a pose increment is not merely $\rho$ when there is a finite rotation; it is coupled through the left Jacobian $J(\phi)$. Conversely, if

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \in \mathrm{SE}(3), \tag{1.18}$$

then the local logarithm is obtained by

$$\phi = \operatorname{Log}(R), \qquad \rho = J(\phi)^{-1}p, \qquad \operatorname{Log}(T) = \begin{bmatrix} \rho \\ \phi \end{bmatrix}. \tag{1.19}$$

Then a pose update is written as

$$T^+ = T\ \operatorname{Exp}(\xi^\wedge). \tag{1.20}$$

The logarithm map does the reverse: it converts a small relative transform into local vector coordinates. For two poses $T_1, T_2 \in \mathrm{SE}(3)$, the relative pose error is

$$\xi = \operatorname{Log}(T_1^{-1}T_2) \in \mathbb{R}^6. \tag{1.21}$$

This is the mathematically correct replacement for "$T_2 - T_1$" when the unknown is a rigid pose. Throughout the report we use this same right-multiplicative convention: nearby poses are represented as $T = \widehat{T}\operatorname{Exp}(\xi^\wedge)$, and local residuals are written with the corresponding right-invariant relative transform.

This machinery cannot be avoided in the present problem. A Gaussian density such as $\mathcal{N}(m, P)$ is defined on a vector space, not on a curved manifold. Therefore the posterior over orientation or pose cannot literally be represented as a Gaussian over the raw entries of $R$, $T$, or a quaternion while still respecting the rotation constraints. The standard construction is to keep the *nominal state* on the manifold but represent uncertainty in local Euclidean coordinates. In other words, the posterior is approximated by

$$T = \widehat{T}\ \operatorname{Exp}(\xi^\wedge), \qquad \xi \sim \mathcal{N}(0, P_T), \tag{1.22}$$

or, for orientation alone,

$$R = \widehat{R}\ \operatorname{Exp}([\delta\theta]_\times), \qquad \delta\theta \sim \mathcal{N}(0, P_R). \tag{1.23}$$

Lie groups are therefore not cosmetic notation. They are what keeps composition, residuals, linearization, and Gaussian uncertainty mutually consistent. The same local-coordinate language reappears in the trajectory prior, in visual residuals, in the EKF error state, and in factor-graph updates [1, 2].

### 1.1.3 Quaternions: storage choice, same local geometry

| **New notation** | **Meaning** |
|---|---|
| $q = [q_w\ q_x\ q_y\ q_z]^\top$ | Quaternion written as one scalar part and a 3D vector part. |
| $q \in \mathbb{S}^3$ | Unit quaternion satisfying $\|q\| = 1$. |
| $q \sim -q$ | Double-cover property: $q$ and $-q$ represent the same physical rotation. |
| $\otimes$ | Quaternion product. |
| $\delta q$ | Small quaternion correction built from a 3D local rotation error. |

Many implementations do not store a rotation matrix directly. They store a unit quaternion

$$q \in \mathbb{S}^3 = \{q \in \mathbb{R}^4 : \|q\| = 1\}. \tag{1.24}$$

Quaternions are a convenient storage format: they are compact, easy to normalize, and easy to propagate numerically. They do *not*, however, turn orientation into an ordinary 4D Euclidean state. A quaternion has a unit-norm constraint and the sign ambiguity $q \sim -q$, since both represent the same element of SO(3).

For that reason, good filters do *not* place a naive 4D Gaussian directly on the quaternion coordinates. They follow exactly the same local-coordinate construction as with rotation matrices: the state stores a nominal quaternion $\widehat{q}$, while uncertainty lives in a 3D rotation error $\delta\theta \in \mathbb{R}^3$. A small correction is converted into a unit quaternion; to first order one may write

$$\delta q \approx \begin{bmatrix} 1 \\ \frac{1}{2}\delta\theta \end{bmatrix} \quad \text{for small } \delta\theta, \tag{1.25}$$

and the nominal quaternion is updated consistently with the right-multiplicative convention above, e.g.

$$q^+ = q \otimes \delta q. \tag{1.26}$$

The quaternion representation is therefore subordinate to the Lie-group story, not a competing alternative to it. One stores orientation by a unit quaternion for convenience, but Bayesian linearization, covariance propagation, and Kalman correction still live in a 3D tangent coordinate. This is the standard multiplicative-quaternion or error-state-quaternion viewpoint [1, 3].

### 1.1.4 Why the unknown is really a random curve

| New notation | Meaning |
|---|---|
| $\bar{x}(t)$ | Continuous-time trajectory state. |
| $\Gamma_T$ | Admissible set of such curves on $[0, T]$. |
| $\Pi_0$ | Prior law on the curve. |

The unknown in trajectory estimation is not a single pose and not a single finite vector. It is a time-indexed object: a curve that assigns a state to every time $t \in [0, T]$. The mathematically clean Bayesian object is therefore a probability measure on a path space. In this report the continuous-time state is written as

$$\bar{x}(t) = \big(R(t), p(t), v(t), \omega(t), b_g(t), b_a(t)\big), \tag{1.27}$$

where orientation is manifold-valued and the remaining variables are Euclidean. A convenient admissible path space is

$$\begin{aligned}\Gamma_T = \Big\{\bar{x} : [0, T] \to \mathrm{SO}(3) \times (\mathbb{R}^3)^5 \;\Big|\; & v, \omega, b_g, b_a \in C([0, T]), \\ & p, R \text{ are absolutely continuous,} \\ & \dot{p} = v,\ \dot{R} = R[\omega]_\times \text{ a.e.}\Big\}.\end{aligned} \tag{1.28}$$

The exact function-space choice is not the main point. The point is that the prior $\Pi_0$ is defined on whole curves and encodes which trajectories are plausible before any data are seen [4, 5].

## 1.2 Problem setup and measurement interface

| New notation | Meaning |
|---|---|
| $W, B, C$ | World, body/IMU, and camera coordinate frames. |
| $e_n$ | Event label in chronological order. |
| $\tau_n$ | Timestamp of event $e_n$. |
| $s_n \in \{\mathrm{imu}, \mathrm{cam}\}$ | Event type: IMU packet or camera frame. |
| $u_n$ | IMU packet at event $n$, containing measured angular rate and specific force. |
| $Y_k$ | Raw image-side data at frame $k$, i.e. detected tag corners and quality metadata. |
| $p_k^-(x_k, M)$ | Predictive joint distribution of current body state and static map at camera time $t_k$. |
| $\lambda_k$ | Prior-free visual likelihood / information message extracted from camera inversion. |
| $z_k^v$ | Reduced fallback pose pseudo-measurement, with $z_k^v = (\widehat{T}_{WC,k}, R_{v,k})$. |
| $\mathcal{J}_{\mathrm{aux}}$ | Index set of non-anchor tags whose world poses are unknown but stationary. |
| $M$ | Static auxiliary-tag map $M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\mathrm{aux}}}$. |
| $T_{BC}$ | Known rigid transform from IMU/body frame to camera frame. |

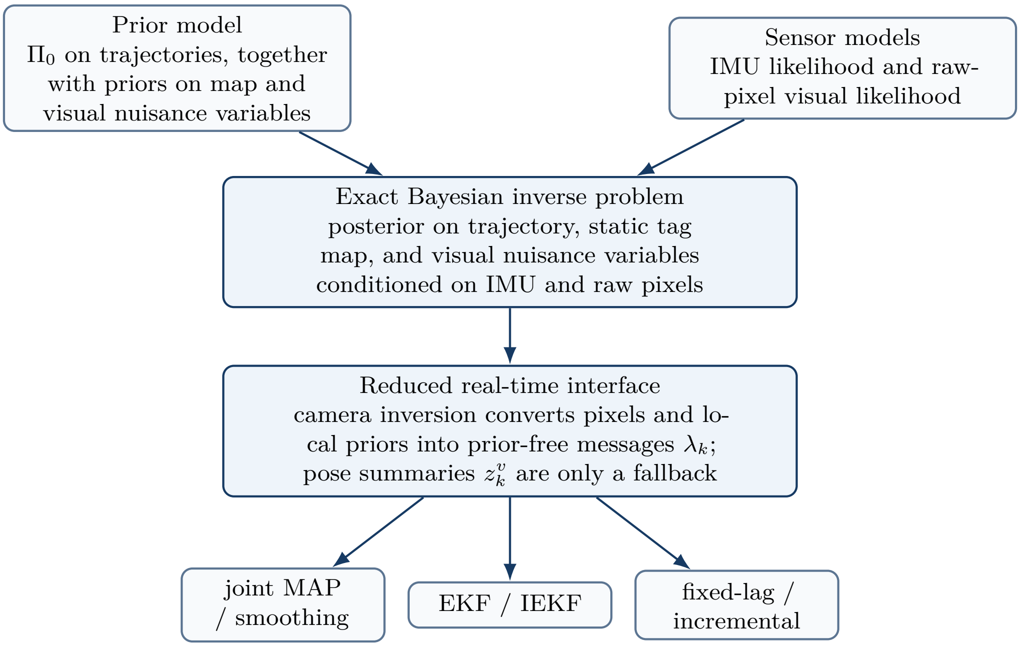


**Figure 1.1:** Conceptual hierarchy. The reference model is a posterior on trajectory, map, and nuisance variables; the real-time estimator works with condensed front-end summaries and then approximates that reduced problem numerically.

We assume a rigid body carrying an IMU and a camera. The world frame $W$ is anchored by at least one known calibration pattern. In the multi-tag case emphasized later, tag $j = 0$ is the anchor and its world pose $T_{WT_0}$ is known (often chosen as the identity pose), while the remaining tags $j \in \mathcal{J}_{\text{aux}}$ are only known to be stationary. The IMU/body frame is $B$, the camera frame is $C$, and the rigid body-to-camera transform

$$T_{BC} = \begin{bmatrix} R_{BC} & t_{BC} \\ 0 & 1 \end{bmatrix} \in \mathrm{SE}(3) \tag{1.29}$$

is assumed known from an earlier calibration step.

The sensor stream is asynchronous. Measurements are assimilated one event at a time, not in artificially synchronized camera–IMU pairs. Let

$$0 \le \tau_1 < \tau_2 < \cdots < \tau_N, \qquad s_n \in \{\text{imu}, \text{cam}\}, \tag{1.30}$$

and let $\mathcal{D}_{1:n}$ denote all sensor events observed up to event $e_n$. If $s_n = \text{imu}$, the event carries an IMU packet

$$u_n = (\widetilde{\omega}_n, \widetilde{f}_n, \tau_n), \tag{1.31}$$

where $\widetilde{\omega}_n$ is measured angular velocity and $\widetilde{f}_n$ is measured specific force. If $s_n = \text{cam}$, the event is a camera frame acquired at time $t_k = \tau_n$ with raw visual data

$$Y_k = \{\widetilde{u}_{i,k}^{(j)},\ s_{i,k}^{(j)},\ \text{tag id } j\}_{j \in \mathcal{V}_k,\ i=1,\dots,m_j}, \tag{1.32}$$

where $\widetilde{u}_{i,k}^{(j)} \in \mathbb{R}^2$ is the detected pixel coordinate of corner $i$ of tag $j$, $s_{i,k}^{(j)}$ denotes detector quality metadata such as margin or score, and $\mathcal{V}_k$ is the set of visible tags at frame $k$. In the simplest anchor-only case the front-end computes

$$x(t) = \big(R_{WB}(t), p_{WB}(t), v_W(t), b_g(t), b_a(t)\big),$$

and the static map is

$$M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}}.$$

The anchor pose $T_{WT_0}$ is fixed and known. The exact Bayesian target is therefore a joint posterior over the dynamic body trajectory and the static auxiliary-tag map conditioned on the full event stream.

The central modeling distinction in this report is between three objects:

1. the predictive prior at a camera time, $p_k^-(x_k, M) = p(x(t_k), M \mid \mathcal{D}_{<k})$,

2. the raw pixel likelihood $p(Y_k \mid x_k, M)$,
3. the posterior after the camera event, $p_k^+(x_k, M) \propto p(Y_k \mid x_k, M)\, p_k^-(x_k, M)$.

This distinction matters because a modular camera subproblem may use the predictive prior internally. If it returns a posterior and the main estimator then treats that posterior as a new independent measurement, the prior has been counted twice. The defensible modular interface is therefore a *prior-free visual message* $\lambda_k$, or equivalently an information increment $(\Delta\Lambda_k, \Delta\eta_k)$ extracted from the camera inversion. A reduced pose pseudo-measurement

$$z_k^v = (\widehat{T}_{WC,k}, R_{v,k}) \tag{1.33}$$

is retained later as a weaker fallback approximation, useful only when one accepts that pose–map correlation and prior reuse are being simplified away.

This event-driven viewpoint keeps the modeling honest while still allowing modular implementations. The camera inversion layer is free to use nonlinear optimization, robust losses, sliding windows, or visible-tag subsets internally. The main recursive estimator simply must not confuse the output of that subproblem with a fresh sensor reading unless the prior contribution has been removed first.

### 1.2.1 How the literature instantiates the same inverse problem

| New notation | Meaning |
|---|---|
| Markov prior | Prior that factorizes as $p(x_0)\prod_k p(x_{k+1} \mid x_k)$. |
| GP prior | Gaussian-process prior on the whole continuous-time trajectory curve. |
| MAP | Maximum a posteriori estimate: the minimizer of the negative log posterior. |
| Fixed lag | Approximate smoothing over a moving recent time window. |

The state of the art differs mainly in *how the prior over trajectories is represented* and *how the posterior is approximated.* Path-space Bayesian inversion papers treat the unknown as a random function or path and define the posterior directly on that function space; Stuart gives the general inverse-problem treatment, and Apte et al. make the path-space viewpoint explicit for data assimilation [4, 5]. Classical recursive Bayesian tracking assumes a discrete-time Markov prior and performs prediction and update sequentially; EKF, UKF, and particle filters are all approximations to this same recursive posterior [6, 7]. Filter-based visual–inertial odometry, exemplified by MSCKF and related consistency-focused filter formulations, uses recursive filtering with inertial propagation and visual correction [8–10]. Optimization-based methods such as OKVIS, IMU preintegration methods, and iSAM2 represent the posterior through a negative log posterior and solve it by sliding-window or incremental nonlinear optimization [11–13]. Continuous-time trajectory methods use GP priors generated by stochastic differential equations and then perform batch GP regression / smoothing on the curve [14, 15].

## 1.3 Bayesian model for the trajectory prior

| New notation | Meaning |
|---|---|
| $\bar{x}(t)$ | Continuous-time latent state. In the prior model we use $\bar{x}(t) = (R(t), p(t), v(t), \omega(t), b_g(t), b_a(t))$. |
| $w_v, w_\omega$ | White-noise drives for translational and rotational smoothness. |
| $w_{bg}, w_{ba}$ | White-noise drives for gyro and accelerometer bias drift. |
| $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ | Spectral-density matrices controlling roughness of the prior. |
| $m_0, P_0$ | Mean and covariance of the initial-state prior. |
| $\Sigma_{\mathrm{map}}$ | Optional prior covariance for a sequence-level visual bias. |

A prior on the curve $\bar{x}(\cdot)$ must say which trajectories are plausible before seeing the data. In this report the prior is introduced as the law of a stochastic process generated by a stochastic differential equation. This section is the first half of the Bayesian model: it explains the trajectory law before any sensor likelihood is applied. Write

$$\bar{x}(t) = \big(R(t), p(t), v(t), \omega(t), b_g(t), b_a(t)\big), \tag{1.34}$$

with kinematic relations

$$\dot{p}(t) = v(t), \qquad \dot{R}(t) = R(t)\,[\omega(t)]_\times\,. \tag{1.35}$$

| Representative work | Unknown object | Prior / regularizer | Inference style | What it assumes or emphasizes |
|---|---|---|---|---|
| Stuart (2010) [4] | Function or curve in infinite-dimensional space | Prior measure on the whole unknown, often Gaussian on a function space | Full Bayesian posterior on the unknown space | General Bayesian inverse viewpoint; emphasizes posterior well-posedness and priors on spaces of functions. |
| Apte et al. (2007) [5] | Continuous-time path | Path-space prior induced by model dynamics and noise | MCMC / path-space Bayesian assimilation | Explicitly formulates data assimilation as conditioning a probability law on paths. |
| Arulampalam et al. (2002) [6] | Discrete state trajectory | Markov state-transition prior | Recursive Bayesian filtering, including particle filters | Generic nonlinear / non-Gaussian online Bayesian tracking. |
| Mourikis–Roumeliotis (2007) [8] | Pose, velocity, biases, camera clones | IMU-driven Gaussian Markov prior with bias random walks | EKF / MSCKF | Real-time vision-aided inertial navigation with recursive updates. |
| Leutenegger et al. (2015) [11] | Keyframe poses, velocities, biases, landmarks | Gaussian priors plus IMU factors and robust visual residuals | Nonlinear MAP / fixed-lag optimization | Tightly coupled VIO with sliding-window refinement. |
| Forster et al. (2017) [12] | Keyframe states plus preintegrated increments | Discrete Gaussian motion prior expressed by on-manifold IMU factors | MAP / factor graph / incremental smoothing | Efficient manifold-correct inertial factors and delayed bias correction. |
| Anderson–Barfoot–Tong–Särkkä (2015) [14] | Entire continuous-time trajectory curve | GP prior generated by a linear time-varying SDE | Batch GP regression / smoothing | Continuous-time prior, interpolation at arbitrary times, exact sparsity. |

**Table 1.2:** Representative Bayesian formulations for trajectory-pose estimation. The same problem can be expressed through path-space priors, discrete Markov priors, MAP objectives, or continuous-time Gaussian-process priors.

A smoothness prior is then generated by

$$\dot{v}(t) = w_v(t), \qquad w_v(t) \sim \mathcal{N}(0, Q_v\, \delta(t-t')), \tag{1.36}$$
$$\dot{\omega}(t) = w_\omega(t), \qquad w_\omega(t) \sim \mathcal{N}(0, Q_\omega\, \delta(t-t')), \tag{1.37}$$
$$\dot{b}_g(t) = w_{bg}(t), \qquad w_{bg}(t) \sim \mathcal{N}(0, Q_{bg}\, \delta(t-t')), \tag{1.38}$$
$$\dot{b}_a(t) = w_{ba}(t), \qquad w_{ba}(t) \sim \mathcal{N}(0, Q_{ba}\, \delta(t-t')). \tag{1.39}$$

Under the continuous-time white-noise idealization, these equations are interpreted in integral form and define a probability law $\Pi_0 = \mathrm{Law}(\bar{x})$ on $\Gamma_T$. Translational acceleration, angular acceleration, and bias drift are modeled through zero-mean stochastic drives. The matrices $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ are therefore not measurement-noise covariances; they are *hyperparameters of the prior law on trajectories.* They control how rough the trajectory is allowed to be. Small $Q_v$ means the prior expects nearly constant velocity. Large $Q_v$ allows aggressive translational maneuvers. Small $Q_{bg}$ and $Q_{ba}$ imply nearly constant biases, while larger values allow faster drift.

A commonly used formal quadratic action associated with this prior is

$$\mathcal{J}_{\text{prior}}^{\text{formal}}(\bar{x}) = \tfrac{1}{2} \int_0^T \Big( \|\dot{v}\|_{Q_v^{-1}}^2 + \|\dot{\omega}\|_{Q_\omega^{-1}}^2 + \left\|\dot{b}_g\right\|_{Q_{bg}^{-1}}^2 + \left\|\dot{b}_a\right\|_{Q_{ba}^{-1}}^2 \Big) dt, \tag{1.40}$$

subject to the kinematic constraints on $R$ and $p$. Equation (1.40) should be read as a formal shorthand or MAP/discretized surrogate on admissible paths; it is *not* a literal density of $\Pi_0$ with respect to a path-space Lebesgue measure.

### 1.3.1 Why these prior components are reasonable

| New notation | Meaning |
|---|---|
| $Q_v$ | Controls translational roughness. Large values mean rapid changes of acceleration are plausible. |
| $Q_\omega$ | Controls rotational roughness. Large values mean rapid changes of angular velocity are plausible. |
| $Q_{bg}, Q_{ba}$ | Control how quickly gyro and accelerometer biases are allowed to drift. |
| $\rho_p, \rho_R, Q_c$ | Parameters of colored visual-noise state, introduced later. |

The rationale for the prior components is physical. Over short video intervals, position and orientation do not jump discontinuously, so a constant-velocity / constant-angular-velocity prior is a sensible default.

The white-noise drives in (1.39) provide a simple smoothness model that still admits real maneuvers. Bias random walks are used because MEMS bias drift is slow, cumulative, and not known a priori; a random walk is a common tractable model for a slowly varying nuisance offset and integrates well with filters and smoothers. The same bias-random-walk nuisance model is used in Kalibr and in standard visual–inertial formulations such as MSCKF-style filters and preintegrated optimization methods [8, 12, 16].

The prior may also include nuisance terms not tied directly to rigid-body dynamics. A sequence-level visual bias can be represented as

$$b_{\mathrm{map}} \sim \mathcal{N}(0, \Sigma_{\mathrm{map}}), \tag{1.41}$$

where $b_{\mathrm{map}} \in \mathbb{R}^6$ captures slowly varying or constant pose offsets due to imperfect anchor geometry, imperfect camera calibration, or tag-model mismatch. If the anchor and camera model are treated as exact, set $b_{\mathrm{map}} = 0$.

### 1.3.2 From the curve prior to the state-space prior used by a Kalman filter

| New notation | Meaning |
|---|---|
| $\tilde{x}_k$ | Sampled extended state at frame time $t_k$, here $\tilde{x}_k = (R_k, p_k, v_k, \omega_k, b_{g,k}, b_{a,k})$. |
| $x_k$ | Reduced practical filter state, usually $x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k})$ after conditioning on IMU packets. |
| $p(\tilde{x}_{k+1} \mid \tilde{x}_k)$ | One-step Markov transition induced by the discretized SDE. |
| $\theta_{\mathrm{prior}}$ | Collection of prior hyperparameters. |

A Kalman filter does not manipulate the full path measure directly. A sampled SDE is Markov on an extended state that contains all variables needed to close the dynamics. For the prior above that extended state is

$$\tilde{x}_k = (R_k, p_k, v_k, \omega_k, b_{g,k}, b_{a,k}), \tag{1.42}$$

and its sampled law factorizes as

$$p(\tilde{x}_{0:N}) = p(\tilde{x}_0) \prod_{k=0}^{N-1} p(\tilde{x}_{k+1} \mid \tilde{x}_k; \theta_{\mathrm{prior}}), \tag{1.43}$$

and hyperparameter set

$$\theta_{\mathrm{prior}} = \{m_0, P_0, Q_v, Q_\omega, Q_{bg}, Q_{ba}, \Sigma_{\mathrm{map}}, \rho_p, \rho_R, Q_c\}. \tag{1.44}$$

In many practical VIO filters the angular velocity is not carried as a separate state because the gyroscope packets are treated as observed inputs in the propagation step. After conditioning on the IMU packets between frames, one obtains the reduced transition kernel $p(x_{k+1} \mid x_k, \mathcal{U}_k)$ for

$$x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k}). \tag{1.45}$$

Thus the Markov prior used by a filter is not a different physical prior; it is a sampled and, if desired, measurement-conditioned form of the same prior-on-curves.

A convenient initial prior is specified locally around a nominal initial state $m_0$: rotational components are perturbed multiplicatively, Euclidean components additively, and the local error covariance is $P_0$. The nominal state $m_0$ is usually derived from the first reliable visual pose, a short stationary IMU segment, or trusted setup geometry, and $P_0$ expresses how uncertain that initialization is. A common block structure is

$$P_0 = \mathrm{diag}(\sigma_{R0}^2 I_3, \sigma_{p0}^2 I_3, \sigma_{v0}^2 I_3, \sigma_{bg0}^2 I_3, \sigma_{ba0}^2 I_3). \tag{1.46}$$

### 1.3.3 Prior for stationary but initially unknown auxiliary tags

| New notation | Meaning |
|---|---|
| $M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\mathrm{aux}}}$ | Static map of auxiliary tag poses. |
| $\pi_M(M)$ | Prior law on the auxiliary-tag map. |
| $\mathcal{W}$ | Coarse workspace region in which auxiliary tags are believed to lie. |
| $\mu_{M,j}, \Sigma_{M,j}$ | Mean and covariance of a local Gaussian prior for auxiliary tag $j$ after anchored initialization. |

When only one tag is anchored, the prior is not only a law on the body trajectory. It also includes a prior on the unknown but stationary auxiliary-tag map

$$M = \{T_{WT_j}\}_{j \in \mathcal{J}_{\text{aux}}}. \tag{1.47}$$

Before any anchored observation of auxiliary tag $j$, the mathematically honest prior is broad: we only know that the tag is stationary and that it lies somewhere in the lab workspace. A minimal prior is therefore a support prior of the form

$$\pi_M(dM) \propto \prod_{j \in \mathcal{J}_{\text{aux}}} \mathbf{1}\{p_{WT_j} \in \mathcal{W}\}\, \pi_{R,j}(dR_{WT_j}), \tag{1.48}$$

where $\mathcal{W} \subset \mathbb{R}^3$ is a coarse workspace box and $\pi_{R,j}$ is an orientation prior reflecting any known mounting convention, for example "tags are roughly vertical on walls" or "tags are roughly horizontal on cube tops."

Once an auxiliary tag is first seen simultaneously with the anchor, one obtains an anchored initialization $\mu_{M,j} = \widehat{T}^{\text{init}}_{WT_j}$. After that point the prior can be tightened and written in local Lie coordinates as

$$T_{WT_j} = \mu_{M,j}\,\mathrm{Exp}(\eta^{\wedge}_{M,j}), \qquad \eta_{M,j} \sim \mathcal{N}(0, \Sigma_{M,j}), \tag{1.49}$$

with small or slowly shrinking covariance $\Sigma_{M,j}$. Again, the Gaussian lives in the local coordinate $\eta_{M,j} \in \mathbb{R}^6$, not directly on SE(3). In a batch or fixed-lag front-end this is the natural way to represent "stationary but initially unknown." In an online filter, one either augments the state with these nearly-static variables or lets a separate front-end map-maintenance module update $\widehat{M}_k$ and its uncertainty.

The key conceptual point is that the anchor removes the global SE(3) gauge freedom, while the stationary auxiliary-tag prior regularizes the remaining map estimation problem. Without the anchor, the camera trajectory and the whole map could drift together by the same rigid transform without changing any pixel prediction.

## 1.4 Bayesian model for the IMU channel

| **New notation** | **Meaning** |
|---|---|
| $\widetilde{\omega}(t)$ | Measured angular velocity. |
| $\widetilde{a}(t)$ | Measured specific force. |
| $g_W$ | Gravity in the world frame. |
| $n_g, n_a$ | Zero-mean gyro and accelerometer white noise. |
| $b_g, b_a$ | Slowly varying gyro and accelerometer biases. |
| $R_g, R_a$ | Continuous-time gyro and accelerometer noise intensities used in the formal likelihood notation; after discretization they contribute to $Q_k$. |

The IMU returns two signals: angular rate from the gyroscope and specific force from the accelerometer. The gyroscope model is

$$\widetilde{\omega}(t) = \omega(t) + b_g(t) + n_g(t), \tag{1.50}$$

where $\omega(t)$ is the true angular velocity expressed in the body frame, $b_g(t)$ is a slowly varying bias, and $n_g(t)$ is fast zero-mean noise.

The accelerometer is more subtle. It does not measure world-frame acceleration directly. It measures the non-gravitational specific force expressed in the sensor frame:

$$\widetilde{a}(t) = R_{BW}(t)\big(\ddot{p}_W(t) - g_W\big) + b_a(t) + n_a(t). \tag{1.51}$$

Here $\ddot{p}_W(t)$ is translational acceleration in the world frame and $g_W$ is the gravity vector in that same frame. The term $R_{BW}(t)(\ddot{p}_W(t) - g_W)$ converts the world-frame specific force into body coordinates. This is why inertial propagation later re-adds gravity in the world frame [12, 17].

### 1.4.1 IMU as a likelihood on the unknown curve

| **New notation** | **Meaning** |
|---|---|
| $\Phi_u(\bar{x}; y^u)$ | IMU negative log-likelihood or data-misfit functional. |
| $\lVert r \rVert^2_{M^{-1}}$ | Weighted squared norm $r^{\mathsf{T}} M^{-1} r$. |

If one stays in the path-space Bayesian language, the IMU contributes a data-misfit functional on the whole curve. Under the common Gaussian white-noise idealization one writes formally

$$\Phi_u(\bar{x}; y^u) = \tfrac{1}{2} \int_0^T \left[ \|\widetilde{\omega}(t) - \omega(t) - b_g(t)\|^2_{R_g^{-1}} + \|\widetilde{a}(t) - R_{BW}(t)(\ddot{p}(t) - g_W) - b_a(t)\|^2_{R_a^{-1}} \right] dt. \tag{1.52}$$

This is the continuous-time analogue of the familiar sum of weighted squared innovations: for any candidate trajectory curve, it quantifies how well that curve explains the measured angular rates and specific forces. As in the prior section, the white-noise notation is formal; rigorous treatments interpret it through discretization or Gaussian-measure constructions rather than through an ordinary Lebesgue density on signal space.

### 1.4.2 IMU as the process model used by a practical online filter

| New notation | Meaning |
|---|---|
| $\mathcal{U}_k$ | All IMU packets between $t_k$ and $t_{k+1}$. |
| $Q_k$ | Effective discrete process covariance over one propagation interval. |

In an online filter it is more convenient to condition on the IMU packets and use the resulting state-transition model. The corresponding continuous-time kinematics are

$$\dot{R}_{WB} = R_{WB} \left[\widetilde{\omega} - b_g - n_g\right]_\times, \tag{1.53}$$

$$\dot{v}_W = g_W + R_{WB}(\widetilde{a} - b_a - n_a), \tag{1.54}$$

$$\dot{p}_{WB} = v_W, \tag{1.55}$$

$$\dot{b}_g = w_{bg}, \tag{1.56}$$

$$\dot{b}_a = w_{ba}. \tag{1.57}$$

A first-order discrete propagation over one camera interval is

$$R_{k+1} = R_k \operatorname{Exp}\big( \left[\widetilde{\omega}_k - b_{g,k}\right]_\times \Delta t\big), \tag{1.58}$$

$$v_{k+1} = v_k + \big(g_W + R_k(\widetilde{a}_k - b_{a,k})\big)\Delta t + n_k^v, \tag{1.59}$$

$$p_{k+1} = p_k + v_k \Delta t + \tfrac{1}{2}\big(g_W + R_k(\widetilde{a}_k - b_{a,k})\big)\Delta t^2 + n_k^p, \tag{1.60}$$

$$b_{g,k+1} = b_{g,k} + n_k^{bg}, \qquad b_{a,k+1} = b_{a,k} + n_k^{ba}. \tag{1.61}$$

The discrete noise terms above are not independent free parameters; they arise from integrating $n_g, n_a, w_{bg}, w_{ba}$ over $[t_k, t_{k+1}]$, and their joint covariance is summarized by $Q_k$. Equation (1.61) is therefore a first-order discretization of the IMU-conditioned state transition used in EKF-, IEKF-, or fixed-lag estimators [2, 8, 12].

## 1.5 Bayesian model for the visual channel

| New notation | Meaning |
|---|---|
| $Y_k$ | Raw visual data at frame $k$: detected corners, tag ids, and quality metadata. |
| $q_k^-, q_k^+$ | Local predictive prior and local posterior used inside a camera inversion subproblem. |
| $\lambda_k$ | Prior-free visual likelihood or information message extracted from the camera inversion. |
| $\Delta\Lambda_k, \Delta\eta_k$ | Information-matrix and information-vector increments representing the visual contribution alone. |
| $z_k^v = (\widehat{T}_{WC,k}, R_{v,k})$ | Reduced fallback pose pseudo-measurement retained for comparison. |
| $P_i^{(j)}$ | Corner $i$ of tag $j$ in tag-local coordinates. |
| $\widetilde{u}_{i,k}^{(j)}$ | Measured image location of a tag corner. |
| $\Sigma_{u,i,k}$ | Pixel-space covariance of one detected corner. |

This section makes the camera-side conditioning structure explicit. At a camera timestamp $t_k$ there are three distinct probabilistic objects:

1. the predictive prior $p_k^-(x_k, M) = p(x(t_k), M \mid \mathcal{D}_{<k})$ supplied by the event-driven estimator,
2. the pixel likelihood $p(Y_k \mid x_k, M)$ supplied by the camera and tag projection model,

3. the posterior $p_k^+(x_k, M) \propto p(Y_k \mid x_k, M)\, p_k^-(x_k, M)$ after the frame is assimilated.

The exact model uses the pixel likelihood directly. A modular camera inversion layer may instead compute a local posterior and then extract a prior-free message $\lambda_k$. A pose pseudo-measurement $z_k^v$ is only a weaker fallback, appropriate when one is willing to discard camera–map correlation and accept a cruder visual interface.

### 1.5.1 Pixel-level camera model

| **New notation** | **Meaning** |
|---|---|
| $K$ | Camera intrinsics matrix. |
| $\pi(K, \cdot)$ | Camera projection function from camera-frame 3D coordinates to pixel coordinates. |
| $T_{WC,k}$ | True camera pose in the world frame. |
| $T_{WT_j}$ | World pose of tag $j$. |
| $u_{i,k}^{(j)}$ | Ideal pixel coordinate predicted by the camera model. |

For a visible tag $j$ and one of its corners $P_i^{(j)}$ in tag-local coordinates, the corresponding world point is

$$P_{W,i}^{(j)} = T_{WT_j} P_i^{(j)}. \tag{1.62}$$

Given the true camera pose $T_{WC,k}$, the ideal pixel prediction is

$$u_{i,k}^{(j)} = \pi\left(K, T_{WC,k}^{-1} P_{W,i}^{(j)}\right) = \pi\left(K, T_{WC,k}^{-1} T_{WT_j} P_i^{(j)}\right). \tag{1.63}$$

The detector returns a noisy corner location

$$\tilde{u}_{i,k}^{(j)} = u_{i,k}^{(j)} + r_{i,k}^{(j)} + d_{i,k}^{(j)} + o_{i,k}^{(j)}, \tag{1.64}$$

where $r_{i,k}^{(j)}$ is the rasterization / quantization floor, $d_{i,k}^{(j)}$ is ordinary corner-localization error from blur and image noise, and $o_{i,k}^{(j)}$ is an occasional larger outlier term due to partial occlusion, decoding failure, or incorrect matching. In a Gaussian approximation one summarizes the inlier part by a covariance $\Sigma_{u,i,k}$ and handles the outlier part by gating or robust losses.

### 1.5.2 Front-end inverse problem when the anchor is known

| **New notation** | **Meaning** |
|---|---|
| $\mathcal{V}_k$ | Set of visible tags in frame $k$. |
| $m_j$ | Number of corners used from tag $j$. |
| $\rho(\cdot)$ | Optional robust loss. |
| $W_k$ | Weight matrix assembled from corner covariances and robust weights. |
| $\xi_{C,k} \in \mathbb{R}^6$ | Local camera-pose perturbation used inside the camera inversion. |

If the world poses of all visible tags are already known, the per-frame camera inversion is a local pose-estimation problem. The direct pixel-side objective is still

$$\widehat{T}_{WC,k} = \underset{T \in \mathrm{SE}(3)}{\operatorname{argmin}} \frac{1}{2} \sum_{j \in \mathcal{V}_k} \sum_{i=1}^{m_j} \rho\left( \left\| \tilde{u}_{i,k}^{(j)} - \pi(K, T^{-1} T_{WT_j} P_i^{(j)}) \right\|_{\Sigma_{u,i,k}^{-1}}^2 \right). \tag{1.65}$$

What changes in the present report is the interface to the main estimator. Let the camera inversion use a local prior

$$q_k^-(\xi_{C,k}) \approx \mathcal{N}(\mu_k^-, P_k^-)$$

derived from the predicted body state at the exposure time. After processing the pixels, it returns a local posterior

$$q_k^+(\xi_{C,k}) \propto p(Y_k \mid \xi_{C,k})\, q_k^-(\xi_{C,k}) \approx \mathcal{N}(\mu_k^+, P_k^+).$$

The correct modular object is then not $q_k^+$ treated as a new measurement. Instead, one removes the local prior in information form:

$$\Delta\Lambda_k = (P_k^+)^{-1} - (P_k^-)^{-1}, \qquad \Delta\eta_k = (P_k^+)^{-1}\mu_k^+ - (P_k^-)^{-1}\mu_k^-. \tag{1.66}$$

The resulting message

$$\lambda_k(\xi_{C,k}) \propto \exp\Big(-\tfrac{1}{2}\xi_{C,k}^{\mathsf{T}}\Delta\Lambda_k\xi_{C,k} + \Delta\eta_k^{\mathsf{T}}\xi_{C,k}\Big)$$

approximates the pixel likelihood contribution alone. Only if the camera inversion is solved with a flat prior, or if one intentionally accepts a weaker reduction, does it collapse to a pose pseudo-measurement summary

$$R_{v,k} \approx (J_k^{\mathsf{T}} W_k J_k)^{-1}, \qquad z_k^v = (\widehat{T}_{WC,k}, R_{v,k}), \tag{1.67}$$

which is then interpreted only as a fallback interface rather than the report's mainline model.

### 1.5.3 Multiple tags when only one tag is anchored

| **New notation** | **Meaning** |
|---|---|
| $T_{WT_0}$ | Known world pose of the anchor tag, often chosen as the identity pose. |
| $\mathcal{J}_{\mathrm{aux}}$ | Set of non-anchor tags. |
| $M = \{T_{WT_j}\}_{j\in\mathcal{J}_{\mathrm{aux}}}$ | Unknown but stationary auxiliary-tag map. |
| $z_k$ | Local variable block containing current camera and visible auxiliary-tag perturbations. |
| $\xi_{T_j} \in \mathbb{R}^6$ | Local perturbation of visible auxiliary tag $j$. |
| $\psi_M(M)$ | Static-map regularizer or prior term. |

The practically important case is that only the anchor tag $j = 0$ has known world pose. All other tags are only known to be stationary. Then the camera event is no longer a pure pose-estimation problem. It is a local joint *pose-plus-visible-map* problem. A compact local variable block is

$$z_k = \big(\xi_{C,k}, \{\xi_{T_j}\}_{j\in\mathcal{V}_k\cap\mathcal{J}_{\mathrm{aux}}}\big),$$

where every pose perturbation lives in local Lie coordinates. The exact camera posterior at time $t_k$ is

$$p_k^+(x_k, M) \propto p(Y_k \mid x_k, M)\, p_k^-(x_k, M), \tag{1.68}$$

and a modular camera inversion computes the corresponding local approximation

$$q_k^+(z_k) \propto p(Y_k \mid z_k)\, q_k^-(z_k).$$

If both are approximated locally by Gaussians,

$$q_k^-(z_k) = \mathcal{N}(\mu_k^-, P_k^-), \qquad q_k^+(z_k) = \mathcal{N}(\mu_k^+, P_k^+),$$

then the message extracted from the frame is again the prior-removed information increment

$$\Delta\Lambda_k = (P_k^+)^{-1} - (P_k^-)^{-1}, \qquad \Delta\eta_k = (P_k^+)^{-1}\mu_k^+ - (P_k^-)^{-1}\mu_k^-, \tag{1.69}$$

now defined on the coupled camera-and-visible-tag block. This is the main reason the message interface is preferable to a bare pose summary: the same pixels constrain current pose and visible tag states together, and the estimator should preserve that local coupling whenever it can.

The weaker alternative is to collapse the camera block to a pose-only summary $z_k^v$. That reduction can still be useful for diagnostics or for a first prototype, but it no longer represents the full visual information carried by jointly visible auxiliary tags, and it becomes unsafe if the summary was built from the same prior already used by the body filter.

### 1.5.4 Why the visual pose covariance changes from frame to frame

| **New notation** | **Meaning** |
|---|---|
| $z_k$ | Approximate range from camera to tag plane. |
| $\theta_k$ | Viewing angle between tag normal and camera optical axis. |
| $A_{\mathrm{tag},k}$ | Projected tag area in pixels. |
| $\beta_k$ | Blur or image-quality penalty. |
| $\sigma_{\mathrm{px},k}$ | Effective corner-localization standard deviation in pixels. |

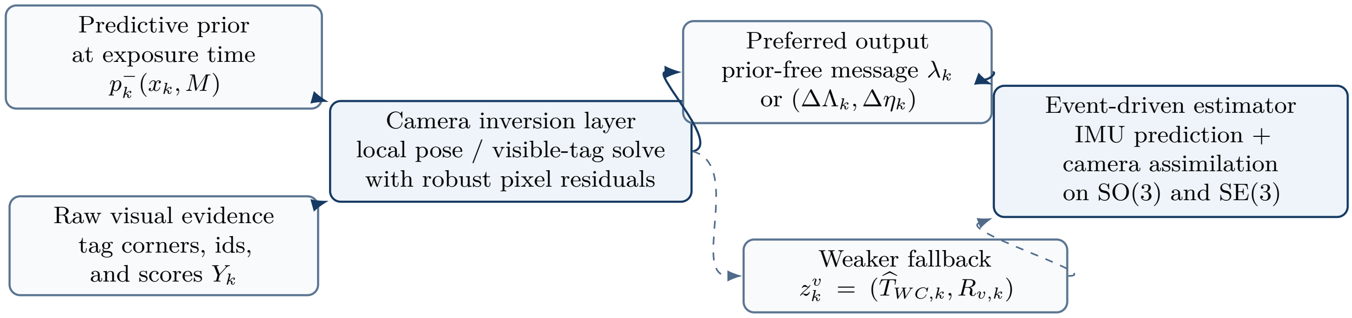


**Figure 1.2:** Reduced visual interface with the recommended message layer. Raw pixels and the predictive local prior enter a camera-specific inversion block. The main estimator should consume the prior-free visual message; a pose summary is retained only as a weaker fallback approximation.

The covariance $R_{v,k}$ is heteroscedastic: it changes with geometry and image quality. The reason is that the underlying PnP-type inverse problem becomes more ill-conditioned when the tag is far away, strongly tilted, small in the image, motion-blurred, or near the image border. A useful engineering model is

$$\sigma_{\mathrm{px},k}^2 = \sigma_0^2 + \frac{\alpha_A}{A_{\mathrm{tag},k}} + \alpha_\theta \tan^2\theta_k + \alpha_b \beta_k^2, \tag{1.70}$$

which says that corner localization gets worse when the tag covers fewer pixels, becomes more oblique, or is more blurred. Local PnP geometry then implies approximate scaling laws such as

$$\sigma_{xy,k} \propto \frac{z_k}{f}\sigma_{\mathrm{px},k}, \qquad \sigma_{z,k} \propto \frac{z_k^2}{f\,\ell_{\mathrm{eff},k}}\sigma_{\mathrm{px},k}, \qquad \sigma_{\mathrm{rot},k} \propto \frac{z_k}{f\,\ell_{\mathrm{eff},k}}\sigma_{\mathrm{px},k}, \tag{1.71}$$

where $f$ is focal length in pixels and $\ell_{\mathrm{eff},k}$ is the effective informative tag span in the image. Experimental AprilTag and planar-fiducial studies report that distance and obliquity degrade pose accuracy [18–20].

In the event-driven message formulation, these same geometry effects determine the local Hessian of the camera inversion and therefore the strength of the information increment $(\Delta\Lambda_k, \Delta\eta_k)$. If one falls back to a pose pseudo-measurement, they determine $R_{v,k}$. In both cases the visual contribution is heteroscedastic and must not be modeled as one fixed covariance for all frames.

### 1.5.5 Temporal correlation in reduced visual summaries

| New notation | Meaning |
|---|---|
| $c_k$ | Temporally correlated component of visual pose error. |
| $b_{\mathrm{map}}$ | Sequence-level static visual bias term. |
| $\Phi, Q_c$ | State-transition matrix and process covariance of the colored visual-noise state. |

Adjacent reduced visual summaries are often correlated because the same tags, the same camera calibration, the same optical conditions, and the same front-end initialization are reused across many consecutive frames. A useful model for the weaker pose-summary interface is therefore

$$\mathrm{Log}\Big(\widehat{T}_{WC,k}^{-1} T_{WC,k}\Big) = b_{\mathrm{map}} + c_k + \nu_k, \qquad \nu_k \sim \mathcal{N}(0, R_{v,k}), \tag{1.72}$$

with colored term

$$c_{k+1} = \Phi c_k + w_k^c, \qquad \Phi = \mathrm{diag}(\rho_p I_3, \rho_R I_3), \qquad w_k^c \sim \mathcal{N}(0, Q_c). \tag{1.73}$$

The constant term $b_{\mathrm{map}}$ captures slowly varying or sequence-level bias, while $c_k$ captures short-memory temporal correlation. If one ignores these terms and pretends that the reduced visual summaries are i.i.d. Gaussian, the resulting filter is usually overconfident. Even the message interface can inherit mild correlation if successive camera subproblems share a sliding window or stale map linearization, so any reduced interface should be described as an approximation rather than as exact Bayes.

## 1.6 Inference structure: exact posterior and practical approximations

| **New notation** | **Meaning** |
|---|---|
| $\Pi(d\bar{x}, dM, dc, db_{\text{map}} \mid \mathcal{D})$ | Full posterior measure on trajectory, map, and visual nuisance variables conditioned on the event stream. |
| $\Pi_{\text{ref}}$ | Product reference prior measure $\Pi_0 \otimes \pi_M \otimes \Pi_c \otimes \pi_{\text{map}}$. |
| $\psi_n$ | Event factor: inertial transition likelihood or camera likelihood contribution. |
| $\widetilde{\Pi}(d\bar{x}, dM, dc, db_{\text{map}} \mid y^u, \lambda)$ | Reduced approximate posterior after replacing each camera frame by a prior-free message. |
| $\Phi(\cdot)$ | Negative log likelihood or data-misfit functional. |
| $J(\cdot)$ | Discretized negative log posterior / MAP objective. |
| $A_k$ | Selector that injects a local camera message into the relevant joint-state coordinates. |

At the fully Bayesian level there are four unknown objects: the continuous trajectory curve $\bar{x}(\cdot)$, the stationary auxiliary-tag map $M$, the temporally correlated visual-error process $c$, and the sequence-level visual bias $b_{\text{map}}$. The clean posterior on these unknowns is a probability measure. If $y^u$ denotes the raw IMU stream and $Y$ the raw visual stream, then Bayes' formula is written symbolically as

$$\Pi(d\bar{x}, dM, dc, db_{\text{map}} \mid y^u, Y) \propto p(y^u \mid \bar{x})\, p(Y \mid \bar{x}, M, c, b_{\text{map}})\, \Pi_0(d\bar{x})\, \pi_M(dM)\, \Pi_c(dc)\, \pi_{\text{map}}(db_{\text{map}}). \quad (1.74)$$

This is the most faithful statement of the inverse problem. It says: start from a prior law on trajectories and nuisance variables, then condition that law on the observed data. To make the Radon–Nikodym formula precise, introduce the product reference prior measure

$$\Pi_{\text{ref}} = \Pi_0 \otimes \pi_M \otimes \Pi_c \otimes \pi_{\text{map}}.$$

Then one may write

$$\frac{d\Pi}{d\Pi_{\text{ref}}}(\bar{x}, M, c, b_{\text{map}}) \propto \exp\big( - \Phi_u(\bar{x}; y^u) - \Phi_v(\bar{x}, M, c, b_{\text{map}}; Y)\big). \quad (1.75)$$

This notation emphasizes that the posterior is still a measure on functions, not yet a finite-dimensional Gaussian vector [4, 5].

After time discretization, the same model can be written as an event-factorized posterior. If $x_n$ denotes the dynamic state at event time $\tau_n$, then

$$p(x_{1:N}, M, c, b_{\text{map}} \mid \mathcal{D}_{1:N}) \propto p(x_1)p(M)p(c_1, b_{\text{map}}) \prod_{n=2}^{N} \psi_n, \quad (1.76)$$

with event factor

$$\psi_n = \begin{cases} p(u_n \mid x_n, x_{n-1}), & s_n = \text{imu}, \\ p(Y_k \mid x_n, M, c_n, b_{\text{map}}), & s_n = \text{cam}. \end{cases}$$

Equation (1.76) is the discrete counterpart of the path-space posterior: each physical data source contributes exactly one factor.

In the preferred reduced real-time formulation the raw pixels are *not* replaced by a camera posterior. They are replaced, frame by frame, by prior-free camera messages $\lambda_k$. The reduced approximate posterior is therefore

$$\widetilde{\Pi}(d\bar{x}, dM, dc, db_{\text{map}} \mid y^u, \lambda) \propto p(y^u \mid \bar{x}) \prod_k \lambda_k(\bar{x}, M, c, b_{\text{map}})\, \Pi_0(d\bar{x})\, \pi_M(dM)\, \Pi_c(dc)\, \pi_{\text{map}}(db_{\text{map}}). \quad (1.77)$$

Here each $\lambda_k$ is meant to approximate the visual likelihood contribution of frame $k$ alone. The weakest reduction replaces $\lambda_k$ by a pose pseudo-measurement kernel on $z_k^v$, but that step is an additional approximation rather than the report's mainline model.

### 1.6.1 Three numerical routes from the posterior to an estimator

| **New notation** | **Meaning** |
|---|---|
| MCMC / SMC | Sampling-based approximation of the posterior law. |
| MAP | Maximum a posteriori estimate obtained by minimizing the negative log posterior. |
| Fixed lag | Sliding-window approximation that repeatedly solves a recent MAP problem. |
| EKF / IEKF | Recursive Gaussian approximation to the filtering posterior. |

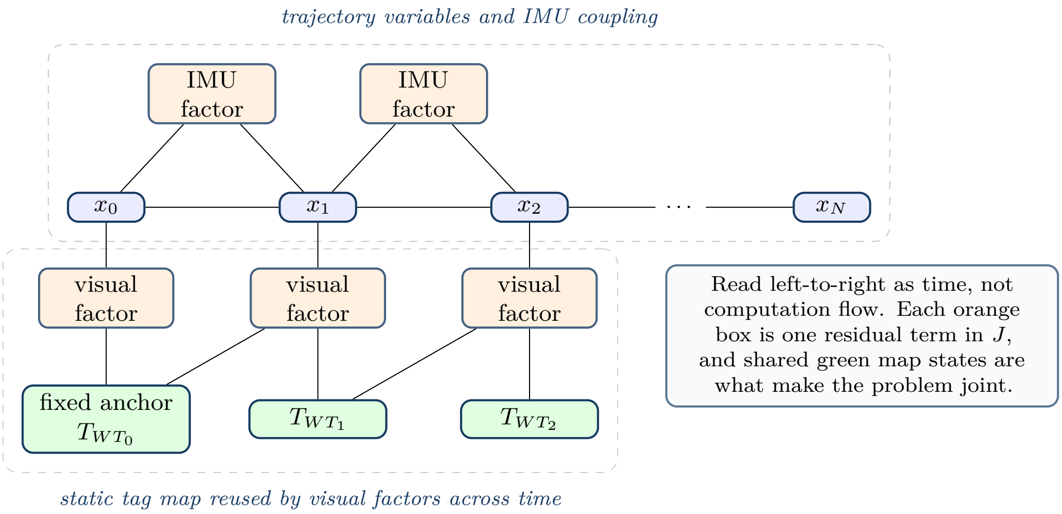


**Figure 1.3:** Sparsity pattern of the exact joint MAP problem. Body states $x_k$ and static tag poses $T_{WT_j}$ are optimized together; IMU factors couple consecutive states, while visual factors couple each frame state to the anchor or auxiliary tags visible in that frame. The visual-correlation variables $c$ and $b_{\text{map}}$ are omitted for readability.

There are three standard ways to turn the Bayesian inverse problem into a numerical algorithm.

**Route A: sample the posterior.** One can approximate the full posterior in (1.74) by Markov-chain Monte Carlo or by sequential Monte Carlo. This is closest to the mathematically exact Bayesian formulation and best reflects multimodality or non-Gaussian uncertainty, but it is usually too expensive for 60 Hz online pose tracking.

**Route B: compute a MAP estimate.** After choosing a finite-dimensional discretization and local coordinates, one obtains a cost functional of the form

$$J(X, M, c, b_{\text{map}}) = \Phi_u(X; U) + \Phi_v(X, M, c, b_{\text{map}}; Y) + J_{\text{prior}}(X, M, c, b_{\text{map}}), \tag{1.78}$$

and then solves

$$(X^\star, M^\star, c^\star, b^\star_{\text{map}}) = \operatorname{argmin} J. \tag{1.79}$$

Here $J_{\text{prior}}$ collects the negative log prior terms in the chosen local coordinates. This becomes the factor-graph / bundle-adjustment view used by fixed-lag smoothers and modern VIO systems [11–13]. Figure 1.3 should be read as the sparsity pattern of that exact joint MAP problem after discretizing the trajectory: blue nodes are unknown body states, green nodes are static tag poses, orange boxes are residual factors, and an edge means that residual term depends on that variable. It is therefore a coupling graph, not a process-flow diagram.

The key structural point is the repeated reuse of the same static tag variables. The auxiliary-tag states $T_{WT_1}, T_{WT_2}, \dots$ connect to visual factors from multiple frames, so the exact posterior is genuinely joint over trajectory and map. A modular camera-message architecture later in the report is therefore a deliberate approximation to this graph, not the graph itself.

**Route C: approximate the filtering posterior recursively by a Gaussian.** One computes only the sequence of filtering posteriors of a reduced Markov state and represents each by a local Gaussian. In the present report, the preferred recursive route is either a joint raw-residual EKF update or a message-based information update; the pose-summary EKF is retained only as a weaker fallback. In all cases one is approximating the same filtering posterior

$$p(\chi_k, M \mid \mathcal{D}_{0:k}), \tag{1.80}$$

replacing it by a Gaussian in local coordinates around a nominal state. This yields EKF, IEKF, or related recursive filters [2, 7, 8].

### 1.6.2 From the continuous posterior to the discrete Bayesian recursion

| New notation | Meaning |
|---|---|
| $\chi_n$ | Markov state used by the reduced event-level model, typically $\chi_n = (x_n, c_n, b_{\text{map}})$ or an augmentation of it. |
| $f_n$ | Event-time transition kernel induced by inertial dynamics and any retained nuisance-state dynamics. |
| $\lambda_k$ | Prior-free camera message acting on a local subset of the state and map. |
| $z_k^v$ | Pose pseudo-measurement used only in the weakest reduced model. |

To obtain an online estimator, sample the path at event times and build a finite-dimensional Markov state. The extended sampled prior from (1.43) is Markov on $\tilde{x}_n = (R_n, p_n, v_n, \omega_n, b_{g,n}, b_{a,n})$. Practical filters usually condition on the IMU packets and retain the reduced body state

$$x_n = (R_n, p_n, v_n, b_{g,n}, b_{a,n}).$$

If the colored visual error $c_n$ and the static bias $b_{\text{map}}$ are estimated explicitly, define $\chi_n = (x_n, c_n, b_{\text{map}})$; otherwise they are omitted from $\chi_n$ and absorbed into an inflated visual model. The event-driven recursion then has two cases.

**IMU event.** If $s_n = \text{imu}$, propagate from event $n-1$ to event $n$:

$$p^{-}(\chi_n, M) = \int f_n(\chi_n \mid \chi_{n-1}, u_n)\, p(\chi_{n-1}, M \mid \mathcal{D}_{1:n-1})\, d\chi_{n-1}. \tag{1.81}$$

**Camera event.** If $s_n = \text{cam}$ and the corresponding frame index is $k$, then either update directly with the pixel likelihood or, in the preferred modular approximation, with the message $\lambda_k$:

$$p(\chi_n, M \mid \mathcal{D}_{1:n}) \propto \lambda_k(\chi_n, M)\, p^{-}(\chi_n, M). \tag{1.82}$$

Only in the weakest reduction does one replace $\lambda_k$ by a pseudo-measurement kernel $\tilde{g}_k(z_k^v \mid \chi_n; \widehat{M}_k)$. Everything that comes later—particle filters, MAP smoothers, EKF, IEKF—is an approximation either to this event-driven recursion or, one level up, to the original raw-pixel posterior measure.

### 1.6.3 How the Kalman filter is obtained as a simplification

| New notation | Meaning |
|---|---|
| $\hat{X}_n, P_n$ | Nominal joint state and covariance at event time $\tau_n$. |
| $\delta X_n$ | Local error coordinates for the joint dynamic-plus-map state. |
| $F_n, G_n, H_k$ | Jacobians of the transition, noise injection, and direct visual residual models. |
| $\Lambda_n, \eta_n$ | Information matrix and information vector of the Gaussian approximation. |
| $A_k$ | Selector matrix that injects the local camera message into the joint state. |

The Kalman filter is exact only for linear dynamics, linear measurements, and Gaussian noise in a vector space. Our problem violates all three conditions: the process and measurement models are nonlinear, and orientation / pose live on manifolds. The practical route to a Kalman-like algorithm is therefore:

1. choose a discrete event-time state sampled from the continuous curve;
2. propagate a nominal state by the nonlinear inertial model through each IMU event;
3. define a local Euclidean error state in tangent coordinates;
4. linearize the transition and camera-assimilation operators around the current nominal state;
5. approximate the local filtering posterior by a Gaussian in the error coordinates.

This is the error-state EKF / IEKF construction.

For the present problem the nominal dynamic state is

$$\hat{x}_n = (\hat{R}_n, \hat{p}_n, \hat{v}_n, \hat{b}_{g,n}, \hat{b}_{a,n}),$$

and a joint filter augments it by local errors for the actively estimated tag poses. The local error vector is therefore

$$\delta X_n = \begin{bmatrix} \delta\theta_n \\ \delta p_n \\ \delta v_n \\ \delta b_{g,n} \\ \delta b_{a,n} \\ \delta m_{1,n} \\ \vdots \\ \delta m_{J,n} \end{bmatrix} \in \mathbb{R}^{15+6J}. \tag{1.83}$$

If quaternions are used internally, $\delta\theta_n$ is still 3-dimensional; only the stored nominal orientation changes from $\hat{R}_n$ to $\hat{q}_n$. If $c_n$ and/or $b_{\text{map}}$ are estimated explicitly, then $\delta X_n$ is augmented by the corresponding Euclidean error coordinates.

For an IMU event, the linearized prediction step is

$$\delta X_n = F_n \delta X_{n-1} + G_n w_n, \qquad P_n^- = F_n P_{n-1}^+ F_n^\top + G_n Q_n G_n^\top. \tag{1.84}$$

If the static map is part of the same state, its nominal value does not move, but the cross-covariances with the dynamic state do propagate. This is exactly the statistical dependence that a separated pose-summary architecture tends to lose.

At a camera event there are three implementation levels:

1. **Direct raw-residual EKF update.** Linearize the stacked pixel residual itself,

   $$r_k \approx H_k \delta X_n + \nu_k, \qquad \nu_k \sim \mathcal{N}(0, R_k),$$

   and apply the ordinary Joseph-form update

   $$S_k = H_k P_n^- H_k^\top + R_k, \qquad K_k = P_n^- H_k^\top S_k^{-1}, \qquad P_n^+ = (I - K_k H_k) P_n^- (I - K_k H_k)^\top + K_k R_k K_k^\top. \tag{1.85}$$

   This is the most direct recursive approximation to the pixel likelihood.

2. **Preferred modular information update.** If the camera inversion returns a prior-free message on a local variable block, inject it into the joint Gaussian in information form:

   $$\Lambda_n^+ = \Lambda_n^- + A_k^\top \Delta\Lambda_k A_k, \qquad \eta_n^+ = \eta_n^- + A_k^\top \Delta\eta_k, \tag{1.86}$$

   where $A_k$ selects the current camera-pose and visible-tag coordinates touched by the frame. One then converts back to covariance form if needed. This realizes the event-driven KF while preserving the meaning of the camera contribution as a likelihood factor rather than as a recycled posterior.

3. **Fallback pose-summary update.** If one keeps only

   $$h_k(x_k, c_k, b_{\text{map}}) = \operatorname{Log}\Big(\widehat{T}_{WC,k}^{-1} T_{WB,k} T_{BC}\Big) - b_{\text{map}} - c_k, \tag{1.87}$$

   together with covariance $R_{v,k}$, then the update is the familiar pose pseudo-measurement EKF. This is the easiest implementation, but it is lower fidelity and must not be built from a prior-conditioned camera posterior unless that prior has been removed first.

Finally, the corrected local mean is injected back onto the manifold by the same retraction rules used elsewhere in the report:

$$\hat{R}_n^+ = \hat{R}_n^- \operatorname{Exp}([\delta\hat{\theta}_n]_\times), \qquad \hat{p}_n^+ = \hat{p}_n^- + \delta\hat{p}_n,$$

with analogous additive updates for velocity, biases, and any explicitly retained Euclidean nuisance states. Thus the Kalman filter is not a different model from the Bayesian inverse problem. It is a tractable local-Gaussian approximation to the same event-driven posterior, obtained after discretization and linearization.

### 1.6.4 When the Kalman simplification is appropriate and when it is not

| New notation | Meaning |
|---|---|
| Unimodal posterior | A posterior well approximated by one Gaussian bump in local coordinates. |
| Linearization basin | Region in which the current nominal state is close enough to the true state for linearization to be accurate. |
| Reacquisition | First visual correction after a visual dropout interval. |

The EKF / IEKF approximation is appropriate when the posterior remains close to unimodal, the visual updates arrive frequently enough to keep inertial drift under control, and initialization starts inside the basin of attraction of the correct solution. Within that regime, the direct raw-residual update and the prior-free message update are both defensible. The pose-summary fallback is weaker, but can still be serviceable if its covariance is honest and its approximation error is monitored.

The approximation becomes weak when long visual dropouts allow inertial propagation to drift far from the truth, when the camera inversion becomes ambiguous or produces multimodal hypotheses, when outliers are not sufficiently gated, or when the visual approximation is badly misspecified. In such regimes a fixed-lag smoother or a more robust sampling-based approximation is safer. This is why the practical implementation hierarchy later in the report recommends starting from the message-aware event-driven filter and then moving to fixed-lag smoothing when relinearization, delayed correction, or correlation repair becomes important.

# Chapter 2

# Practical Implementation

## 2.1 Parameter and noise guide: prior, likelihood, and tuning roles

| New notation | Meaning |
|---|---|
| $m_0, P_0$ | Mean and covariance of the initial-state prior. |
| $Q_v, Q_\omega$ | Trajectory-smoothness spectral densities in the prior-on-curves. |
| $Q_{bg}, Q_{ba}$ | Bias random-walk spectral densities in the prior. |
| $R_g, R_a$ | IMU likelihood covariance or equivalent noise-density parameters. |
| $\Delta\Lambda_k, \Delta\eta_k$ | Information-form strength and shift of the camera message at frame $k$. |
| $R_{v,k}$ | Fallback pose-summary covariance when a frame is collapsed to $z_k^v$. |
| $\sigma_q, \sigma_0, \alpha_A, \alpha_\theta, \alpha_b$ | Pixel-to-Hessian or pixel-to-pose noise parameters used to build the visual approximation. |
| $\rho_p, \rho_R, Q_c$ | Colored visual-noise hyperparameters for the latent process $c_k$. |
| $\Sigma_{\text{map}}$ | Prior covariance of the sequence-level map / anchor bias term. |

A frequent source of confusion is mixing up *prior parameters* and *likelihood parameters*. The matrices $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ and the initial parameters $m_0, P_0$ define the prior law. They say what kinds of trajectories and bias evolutions are plausible *before* the current measurements are assimilated. By contrast, $R_g, R_a$, the frame-wise camera Hessian, and any derived quantities such as $(\Delta\Lambda_k, \Delta\eta_k)$ or the fallback covariance $R_{v,k}$ describe the spread and information content of the observed measurements around the model-predicted values.

A helpful mental model is:

- prior parameters regularize the *unknown trajectory*;
- likelihood parameters regularize the *mismatch between data and model*;
- nuisance-state priors, such as $Q_c$ or $\Sigma_{\text{map}}$, sit in between and regularize latent error processes that explain persistent discrepancies.

In practice, parameter selection should proceed hierarchically.

1. Estimate $R_g, R_a$ and their continuous-time equivalents from stationary IMU data, preferably through Allan deviation analysis. Kalibr's IMU model uses continuous-time white-noise and bias-random-walk parameters of exactly this form [16].
2. Estimate or model the camera-message strength frame by frame from the local visual Hessian or from a parametric geometry-driven model such as (1.70); if a pose-summary fallback is used, this same step yields $R_{v,k}$.
3. Set $m_0$ from the first trustworthy anchored visual pose and $P_0$ by inflating the corresponding local covariance to reflect unmodeled setup uncertainty.
4. Tune $Q_v, Q_\omega, Q_{bg}, Q_{ba}$ so that innovation statistics are consistent: too small gives under-dispersed posteriors, too large gives noisy or drifting estimates.
5. If residual autocorrelation remains high on the visual side, introduce $c_k$ with parameters $(\rho_p, \rho_R, Q_c)$ rather than forcing independent noise to explain correlated errors.
6. If the camera module internally uses a predicted prior, verify that the exported visual interface is prior-free by construction; otherwise the back-end will be systematically overconfident.

## 2.2 Representative numeric regime for a phone-on-arm lab experiment

| New notation | Meaning |
|---|---|
| $\Delta t_c$ | Camera frame interval. |
| $f_{\text{imu}}$ | IMU sampling rate. |
| $s$ | Tag side length. |
| $z$ | Camera-to-anchor range. |
| $\sigma_g, \sigma_a, \sigma_{bg}, \sigma_{ba}$ | IMU white-noise and bias-random-walk strengths. |

This section gives two kinds of numbers: *hardware facts* drawn from cited sources, and *engineering starting values* for the model parameters used in this chapter. The latter are recommendations, not immutable truths.

| Quantity | Representative value | Rationale / source |
|---|---|---|
| Phone video rate | 60 fps | Google's dedicated Pixel 9a specs page lists rear-camera 4K and 1080p recording at 30/60 fps; taking the 60 fps mode as the working regime gives $\Delta t_c \approx 16.7$ ms for this example [21]. |
| Phone main-camera FoV | about 82° | Google's Pixel hardware tech-specs page lists an 82° field of view for the Pixel 9a wide camera [21]. |
| Phone motion-sensor rate | 200 Hz limit for `registerListener()` on Android 12+ apps without the high-rate permission | Android documents that, for apps targeting Android 12 (API level 31) or higher, `registerListener()` is rate-limited to 200 Hz for the affected motion and position sensors unless the app declares `HIGH_SAMPLING_RATE_SENSORS` [22]. |
| Research tabletop arm reach | about 0.855 m | The Franka Research 3 product page lists an 855 mm reach; this is a realistic scale for anchor-to-workspace geometry in a lab manipulation setup [23]. |
| Robot repeatability | $< \pm 0.1$ mm | The official Franka Research 3 datasheet lists position repeatability below $\pm 0.1$ mm under ISO 9283 [23]. |
| Robot control rate | about 1 kHz low-level control | Official FR3 materials describe 1 kHz control and measurement access through the Franka Control Interface; this is much faster than 60 Hz phone video and faster than the 200 Hz Android sensor-rate limit without the high-rate permission [21–23]. |
| Practical tag sizes | 0.06–0.10 m | Typical lab fiducials for arm-mounted phone experiments are in the 6–10 cm range; this is an engineering choice that keeps tags visible at about 0.4–1.2 m range. |
| Practical anchored range | 0.4–1.2 m | Consistent with a tabletop arm of roughly 0.85 m reach plus camera offset and the need to keep the anchor visible at least at startup. |
| Example Kalibr `imu.yaml` values | $\sigma_a = 1.86 \times 10^{-3}$ $\mathrm{m/s^2/\sqrt{Hz}}$, $\sigma_{ba} = 4.33 \times 10^{-4}$ $\mathrm{m/s^3/\sqrt{Hz}}$, $\sigma_g = 1.87 \times 10^{-4}$ $\mathrm{rad/s/\sqrt{Hz}}$, $\sigma_{bg} = 2.66 \times 10^{-5}$ $\mathrm{rad/s^2/\sqrt{Hz}}$ | Kalibr's YAML-formats page shows these as example `imu.yaml` entries at an update rate of 200 Hz; the units and continuous-time interpretation are documented separately in the Kalibr IMU-noise page, so these values are useful as a format-and-units example, not as a phone-specific default [16, 24]. |

**Table 2.1:** Representative numbers for a phone-on-arm laboratory regime. Some are hardware facts from official specs; others are engineering starting points for model design or sanity-check performance levels.

For a first implementation aimed at a robot arm moving in a lab around an anchored tag, a very reasonable working regime is:

- camera at 60 Hz,
- IMU at 100–200 Hz,
- tag size $s \in [0.06, 0.10]$ m,
- anchor range $z \in [0.4, 1.2]$ m,
- path spans on the order of 0.1–0.5 m in translation over the sequence,
- translational speeds on the order of 0.05–0.5 m/s for careful tabletop motions,
- angular rates on the order of 5°/s to 60°/s for ordinary camera reorientation.

These last values are engineering design targets rather than universal physical constants; they simply place the prior and likelihood parameters in the right numerical regime for a lab arm rather than a UAV or car.

## 2.3 Implementation architecture for the anchored multi-tag problem

| New notation | Meaning |
|---|---|
| $M_k = \{(\widehat{T}_{WT_j,k}, P_{m_j,k})\}_{j\in\mathcal{J}_k}$ | Current map manager state: a set of auxiliary-tag pose posteriors carried up to frame $k$. |
| $P_{m_j,k} \in \mathbb{R}^{6\times6}$ | Local covariance of tag $j$ in tangent coordinates around $\widehat{T}_{WT_j,k} \in \mathrm{SE}(3)$. |
| $\mathcal{J}_k$ | Set of auxiliary tags that have already been initialized by frame $k$. |
| $\mathcal{V}_k^{\mathrm{old}}, \mathcal{V}_k^{\mathrm{new}}$ | Visible tags at frame $k$ that are already in the map, and visible tags observed for the first time. |
| $\lambda_k$ or $(\Delta\Lambda_k, \Delta\eta_k)$ | Prior-free visual message returned by the camera inversion at frame $k$. |
| $z_k^v = (\widehat{T}_{WC,k}, R_{v,k})$ | Weaker fallback pose summary retained for simplified implementations. |

The previous chapter established the reference posterior and the message-based reduced interface. This section translates those objects into an implementation architecture. The central design choice is now:

- If accuracy and statistical correctness are the highest priorities, solve a *joint* posterior over trajectory and static tag map.
- If online speed and modularity are the highest priorities, use an *event-driven message architecture*: the camera module may solve its own local subproblem, but it must export a prior-free visual message to the joint recursive estimator.
- Use a pose pseudo-measurement $z_k^v$ only as a weaker fallback when one explicitly accepts the extra loss of correlation structure.

The rest of the section explains how that architectural choice affects initialization, new-tag birth, reappearance handling, and map bookkeeping. Figure 2.1 should be read as an engineering module diagram for the event-driven modular approximation, not as a posterior graph like Figure 1.3. Solid arrows denote current-event information flow, while the dashed path denotes persistent map memory and cached priors reused by later camera events.

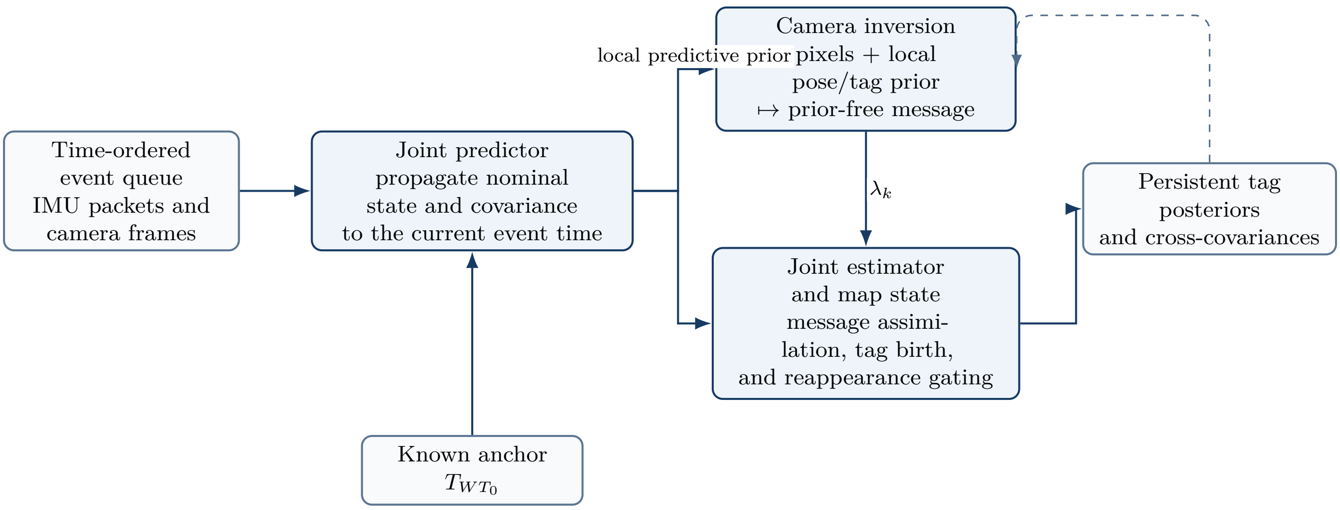


**Figure 2.1:** Event-driven implementation architecture. IMU and camera events are processed in timestamp order. The predictor advances the joint state to the current event time; camera frames trigger a local inversion that returns a prior-free message; the joint estimator assimilates that message while preserving pose–map cross-covariances. Persistent tag beliefs are stored and reused by later camera events.

### 2.3.1 The exact joint solution

The most defensible solution is to keep the entire unknown in one posterior. In discrete time this means solving for

$$X_{0:N} = \{x_0, \dots, x_N\}, \qquad M = \{T_{WT_j}\}_{j\in\mathcal{J}_{\mathrm{aux}}},$$

with posterior density

$$p(X_{0:N}, M \mid U_{0:N-1}, Y_{0:N}) \propto p_0(x_0)p_0(M) \prod_{k=0}^{N-1} p(x_{k+1} \mid x_k, U_{k:k+1}) \prod_{k=0}^{N} p(Y_k \mid x_k, M).$$

No information is used twice: each IMU packet contributes once through the process factor, each corner pixel contributes once through the camera model, and the static map and trajectory are inferred together. In practice this posterior is solved either by fixed-lag smoothing, batch MAP estimation, or posterior sampling as discussed earlier.

The fully joint method is especially attractive when

- anchor visibility is intermittent,
- the auxiliary tags are numerous enough to act as a map,
- one wants statistically honest uncertainty on both camera pose and tag pose,
- and the same data must later support a trajectory posterior, a map posterior, and control diagnostics.

The price is computation and implementation complexity. A real-time version is usually built as a fixed-lag smoother, an EKF-SLAM style filter, or an incremental factor-graph system with a bounded set of actively tracked tags [8, 11, 12, 25].

### 2.3.2 The recommended event-driven message architecture

The recommended real-time alternative is to keep the event-driven timeline but modularize the camera processing. The estimator is then separated into three coupled pieces:

1. a joint recursive estimator that propagates the dynamic state and the active tag-map covariance through IMU events,
2. a camera inversion layer that uses the current local prior and the current pixels to form a camera-only or pose-plus-visible-tag posterior,
3. a message interface that removes the local prior and feeds only the visual likelihood contribution back into the joint estimator.

For IMU events, the joint state is propagated in the usual way. For camera events, the recursion is

$$p_k^-(x_k, M) = p(x(t_k), M \mid \mathcal{D}_{<k}), \tag{2.1}$$

$$q_k^+(z_k) \approx p(Y_k \mid z_k)\, q_k^-(z_k), \tag{2.2}$$

$$\lambda_k(z_k) \propto q_k^+(z_k)/q_k^-(z_k), \tag{2.3}$$

$$p_k^+(x_k, M) \propto \lambda_k(x_k, M)\, p_k^-(x_k, M), \tag{2.4}$$

where the quotient is understood in the local Gaussian or local Laplace sense described in Chapter 1. This architecture preserves the main Bayesian rule: the same prior is not reused twice.

This design is still approximate, because the camera inversion itself is local and usually nonlinear, and because only a subset of the full joint posterior is exposed to the camera block at each frame. But it is materially more faithful than the old pose-summary pipeline. A clean implementation should obey the following rules.

- IMU packets should enter the estimator exactly once, through event-time propagation.
- If the camera inversion uses the predicted pose and tag priors, it must export a prior-free message rather than its posterior.
- Newly seen tags should be born with broad covariance and should not strongly correct global pose on first sighting.
- Reappearing tags should be gated before they are trusted, because stale or moved tags can damage both pose and map.
- Any slower background map refinement or fixed-lag smoother should be described as a posterior-improvement layer, not as a second independent sensor update.

If the message layer is unavailable, a pose pseudo-measurement update remains possible, but it should be labeled honestly as a weaker approximation. That fallback discards the most important camera–map couplings precisely where multi-tag localization is strongest.

Figure 2.2 makes the exposure-time logic explicit. The estimator does not wait for an artificial camera–IMU pair; it propagates through the arriving IMU packets and, when a camera frame arrives, advances the state to that exact exposure time before launching the camera inversion.

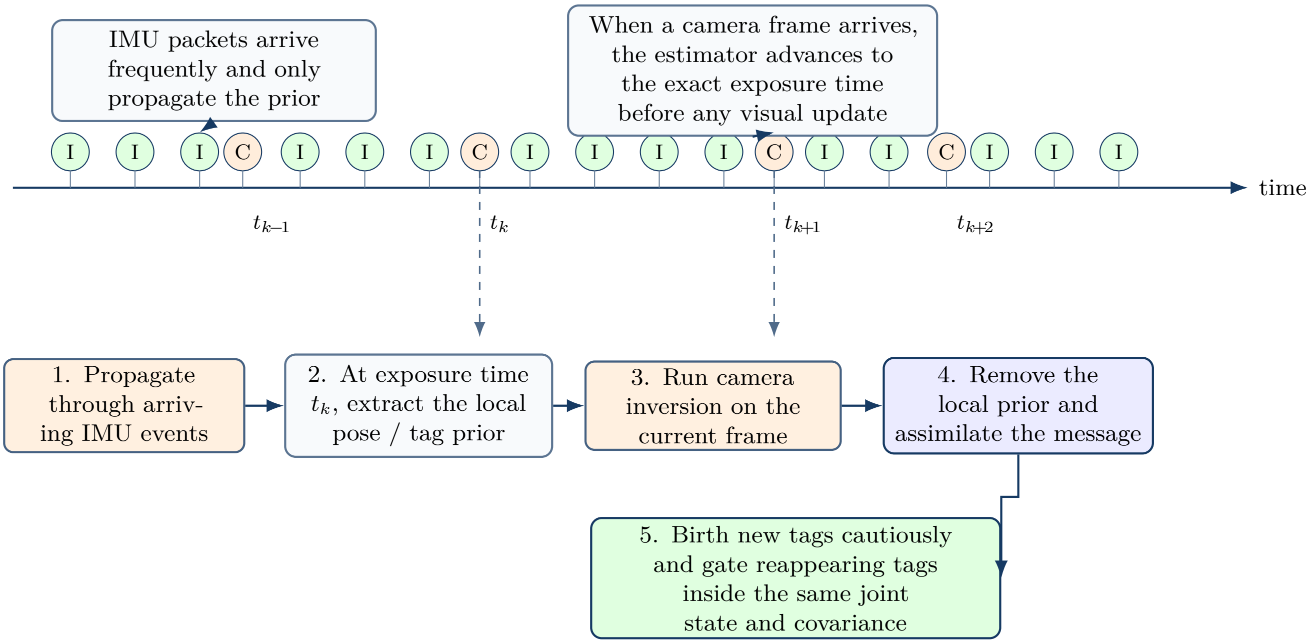


**Figure 2.2:** Exposure-time flow in the recommended modular approximation. Top: asynchronous IMU and camera events are processed strictly in timestamp order. Bottom: when a camera frame arrives at exposure time $t_k$, the estimator first advances to that exact time, extracts the local prior, runs the camera inversion, removes the prior contribution, and assimilates the resulting visual message into the same joint state that stores the active tag posteriors.

### 2.3.3 First frame: anchor guaranteed visible

At frame $k = 0$ the anchor is assumed visible. This makes initialization much simpler because the global gauge is fixed immediately.

**Step 1: initialize the camera pose from the anchor.** Solve the camera inversion using the anchor tag alone, or the anchor plus any already-known tags if such a map exists, to obtain

$$q(T_{WC,0}) \approx \mathcal{N}_{\mathfrak{se}(3)}(\widehat{T}_{WC,0}, R_{v,0}).$$

Then convert to body pose using the known camera–IMU extrinsic,

$$\widehat{T}_{WB,0} = \widehat{T}_{WC,0} T_{CB},$$

and propagate covariance through the local linearization.

**Step 2: initialize non-pose Euclidean components.** Choose

$$v_{W,0} \sim \mathcal{N}(0, P_{v0}), \qquad b_{g,0} \sim \mathcal{N}(0, P_{bg0}), \qquad b_{a,0} \sim \mathcal{N}(0, P_{ba0}),$$

with conservative spreads unless a stationary segment is available.

**Step 3: initialize every auxiliary tag that is already visible.** For each $j \in \mathcal{V}_0 \setminus \{0\}$, condition on the current anchored camera belief and solve a tag-pose inverse problem

$$\widehat{T}_{WT_j,0} = \underset{T \in \mathrm{SE}(3)}{\operatorname{argmin}} \sum_{i=1}^{4} \left\| \widetilde{u}_{i,0}^{(j)} - \pi(K, \widehat{T}_{CW,0} T P_i^{(j)}) \right\|^2_{\Sigma^{-1}_{0,j,i}}. \tag{2.5}$$

Then initialize a broad covariance $P_{m_j,0}$ from the local Hessian, inflated to reflect uncertainty in the camera pose used for conditioning. Intuitively, the anchor gives the first trustworthy camera pose, and that camera pose gives the first absolute estimate of every co-visible auxiliary tag. These newborn tags should be marked provisional and should not dominate global pose correction until they have been reobserved.

### 2.3.4 First sighting of a new tag at a later frame (tag birth)

Suppose tag $j$ is seen for the first time at frame $k > 0$. Then there are two legitimate routes.

**Joint route.** In a joint filter or smoother, augment the state with a new static variable $T_{WT_j}$ and update the enlarged posterior directly using the current pixels. This is the statistically clean route.

**Message-layer route.** In the recommended modular design, use the current body and visible-tag posterior as a conditional prior and create a new tag posterior

$$p(T_{WT_j} \mid Y_k^j, \mathcal{D}_{0:k}) \propto \int p(Y_k^j \mid x_k, T_{WT_j})\, p(x_k \mid \mathcal{D}_{0:k})\, dx_k\, p_0(T_{WT_j}). \tag{2.6}$$

This integral is usually approximated by conditioning on the current body mean or sigma points. A practical implementation is:

1. wait until the current camera/body posterior is well constrained by the anchor or by already-mapped tags,
2. solve (2.5) at the current frame using the current anchored camera mean,
3. assign a broad covariance that includes both pixel uncertainty and the uncertainty in the current camera pose,
4. mark the tag as "provisional" until it has been observed repeatedly.

**Important edge case.** If the camera sees only entirely new unknown tags and no anchor or already-initialized tags, then those tags cannot yet be anchored absolutely in the world frame. In that case the correct action is *not* to let them strongly correct global pose. One should defer firm birth until a frame where the camera itself is well localized in the anchored frame.

### 2.3.5 Tag disappearance and later reappearance

A stationary tag that disappears for several frames does not require a special dynamic model. Its pose remains static:

$$T_{WT_j,k+1} = T_{WT_j,k},$$

possibly with a very small artificial process noise added only for numerical conditioning. Therefore during absence one simply carries its posterior forward,

$$q_{k+1}(T_{WT_j}) = q_k(T_{WT_j}),$$

while recording bookkeeping such as last seen time, number of successful updates, and recent residual quality.

When the tag reappears, one treats the previous tag posterior as the prior and performs an ordinary measurement update. In the joint formulation, the pixels for that tag simply re-enter the global likelihood. In the message-layer formulation, the current body posterior and previous tag posterior produce a predicted pixel distribution, which is then used for gating and update. Formally,

$$p(T_{WT_j} \mid Y_k^j, \mathcal{D}_{0:k}) \propto p(Y_k^j \mid x_k, T_{WT_j})\, p(T_{WT_j} \mid \mathcal{D}_{0:k-1}), \tag{2.7}$$

with the current body posterior either conditioned on jointly or approximated by its mean/covariance.

For robust practice, reappearing tags should be accepted only after a consistency gate:

- predicted-vs-observed corner residual below threshold,
- correct decoded tag id,
- plausible depth and viewing geometry,
- and, if available, Mahalanobis innovation or NIS below a chosen threshold.

If a tag has been absent for a long interval, mild covariance inflation before the update is a defensible safeguard against overconfidence.

### 2.3.6 How to keep track of all tags in code

A simple and effective representation is to store each auxiliary tag as a record

$$m_j = (\widehat{T}_{WT_j}, P_{m_j}, n_j, \tau_j, \eta_j),$$

where $n_j$ is observation count, $\tau_j$ is last-seen timestamp, and $\eta_j$ is a small quality bundle such as mean residual, provisional/confirmed flag, and recent visibility streak. The whole map manager state is then

$$M_k = \{m_j\}_{j \in \mathcal{J}_k}.$$

No global alignment other than the anchor is assumed: every tag carries its own arbitrary 6-DoF pose in the anchored world frame. In a joint filter, these records also imply cross-covariance blocks with the dynamic state. In a slightly more modular implementation, dormant tags can be stored out of the active state and activated only when visible, but the same local Lie-coordinate conventions still apply.

The camera pose itself is tracked in exactly the same geometric space as each tag pose, namely a pose on SE(3) plus a covariance on the corresponding tangent space. The difference is semantic, not mathematical: the camera pose is dynamic and time indexed, while each auxiliary-tag pose is static and time invariant.

### 2.3.7 A compact implementation recipe

For readers who want an executable checklist, the following is a good starting point.

1. **Sort all events by timestamp.** Do not force camera and IMU data into synchronized pairs.
2. **Frame 0.** Use anchor pixels to initialize $T_{WC,0}$, then initialize $T_{WB,0}$, velocity, and biases. Initialize all co-visible auxiliary tags as provisional static poses with broad covariance.
3. **At each IMU event.** Propagate the nominal state and covariance to the event time, preserving cross-covariances with active tags.
4. **At each camera event.** Propagate to the exact exposure time and extract the local prior over current pose and visible known tags.
5. **Run camera inversion.** Use all visible tag pixels and the local priors to build a local posterior.
6. **Remove the local prior.** Export only the prior-free visual message, not the camera posterior itself.
7. **Assimilate jointly.** Update the body state and all visible active tags inside the same covariance or information matrix.
8. **Birth new tags cautiously.** Initialize them with broad covariance and prevent them from strongly correcting global pose until reobserved.
9. **Gate reappearing tags.** Use residual and NIS checks before accepting them into the update.
10. **Bookkeeping.** Increment counts, store residual statistics, mark tags as confirmed after enough successful reobservations, and keep absent tags unchanged until they reappear.

This recipe keeps the module boundaries clean while preserving the key Bayesian rule that each physical frame contributes one visual likelihood factor exactly once.

## 2.4 Kalman realization, sampling, and approximation pitfalls

| New notation | Meaning |
|---|---|
| $\pi(\cdot)$ | Generic posterior density. |
| $\theta$ | Collective variable used in sampling, e.g. a block of poses or tag states. |
| $\alpha(\theta, \theta')$ | Metropolis–Hastings acceptance probability for a proposal $\theta \to \theta'$. |
| $\Lambda, \eta$ | Information matrix and information vector of the local Gaussian approximation. |
| NIS, NEES | Innovation and state consistency statistics used to validate the Gaussian approximation. |

### 2.4.1 How the Kalman filter emerges from the Bayesian problem

The exact filtering recursion was written in (1.82). A practical Kalman-style realization appears only after several simplifications:

1. discretize the curve prior into a Markov transition law,
2. represent the posterior by a local Gaussian in tangent coordinates,

3. linearize the process and measurement operators,
4. assume that the necessary marginal or conditional distributions are adequately captured by their first two moments.

The implementation consequence is that the filter should carry the current dynamic state and any active tag states in one Gaussian object. If the covariance is partitioned as

$$P_n^- = \begin{bmatrix} P_{xx,n}^- & P_{xM,n}^- \\ P_{Mx,n}^- & P_{MM,n}^- \end{bmatrix},$$

then an IMU propagation updates not only the dynamic block but also the cross-covariances:

$$\bar{x}_n = f(\widehat{x}_{n-1}, u_n), \qquad \bar{P}_{xx,n} = F_n P_{xx,n-1} F_n^\top + G_n Q_n G_n^\top, \qquad \bar{P}_{xM,n} = F_n P_{xM,n-1}.$$

That line is easy to miss and is exactly why separated tag-map updates tend to become overconfident.

At a camera event, the preferred reduced implementation is the information-form message update from (1.86). One keeps

$$\Lambda_n^- = (P_n^-)^{-1}, \qquad \eta_n^- = \Lambda_n^- \mu_n^-,$$

adds the message, and converts back to covariance form only if needed. This is the best modular compromise: the camera code stays self-contained, but the estimator still treats the frame as one likelihood contribution.

The raw-residual EKF update is formally cleaner and should be preferred whenever the codebase can tolerate direct camera residuals inside the estimator. The pose pseudo-measurement update is the easiest engineering shortcut, but it should be treated as a downgrade in statistical fidelity rather than as the canonical Kalman realization.

### 2.4.2 Metropolis–Hastings sampling for posterior visualization

For scientific understanding and posterior visualization, it is useful to sample not only trajectories but also tag poses. Let

$$\theta = (X_{0:N}, M)$$

or, for a fixed-lag window, a smaller block such as recent poses, recent biases, and the currently visible tag poses. Given a proposal density $q(\theta' \mid \theta)$, the Metropolis–Hastings acceptance probability is

$$\alpha(\theta, \theta') = \min\left\{1, \frac{\pi(\theta')\, q(\theta \mid \theta')}{\pi(\theta)\, q(\theta' \mid \theta)}\right\}. \tag{2.8}$$

A practical choice is blockwise proposals in local coordinates:

- propose several consecutive body poses in $\mathfrak{se}(3)$ coordinates,
- propose velocity and bias blocks additively in $\mathbb{R}^3$,
- propose one or more tag poses in $\mathfrak{se}(3)$ coordinates.

This is not meant for real-time estimation at 60 Hz. Its purpose is diagnostic and pedagogical: it reveals multimodality, non-Gaussianity, and tag–trajectory coupling that a local EKF or message-linearization approximation may hide. For example, one can visualize posterior samples of a reappearing tag's pose, or posterior tubes of the camera trajectory during partial occlusion.

### 2.4.3 What counts as a statistical or probabilistic crime?

Several implementation shortcuts are acceptable approximations. Several others are genuine mistakes.

- **Reusing a camera posterior as a measurement.** If the camera inversion uses the predicted pose prior and its posterior is fed back as if it were an independent sensor reading, the prior has been counted twice.
- **Double counting IMU.** If IMU is used inside the camera module to form a visual summary and then used again in the back-end process model, inertial information has been counted twice.
- **Using the same pixels twice as if independent.** If a frame's pixels are first condensed into a posterior and then used again to update the same latent variables without accounting for dependence, posterior uncertainty becomes too small.

- **Birth of a new tag without an anchored camera posterior.** A tag seen in isolation cannot be assigned a trustworthy absolute world pose unless the camera pose in the anchored frame is already known.
- **Treating tag poses as Euclidean vectors.** Tag pose covariance must live in local coordinates on SE(3), not in arbitrary matrix-entry coordinates.
- **Ignoring time correlation in reduced visual summaries.** If the front-end reuses similar geometry frame after frame, assuming independent $R_{v,k}$ may lead to overconfident trajectory estimates.

On the other hand, the following are usually acceptable first approximations:

- using the current camera mean to initialize a new tag and inflating its covariance,
- keeping a slower background map-refinement module or fixed-lag smoother for dormant tags,
- using a fixed-lag smoother to repair the neglected correlations of the real-time front-end.

### 2.4.4 Assessment: is the modular message scheme reasonable?

Yes, with one correction: the recommended reduced architecture is no longer the old staggered pose-summary pipeline. The reasonable modular scheme is the event-driven message architecture described above. It is not the exact joint posterior of Chapter 1, but it is a principled approximation of it. The camera code can stay separate, the back-end can remain recursive and real-time, and a slower map-refinement layer can still be added later. What is no longer acceptable as the default story is to treat a prior-conditioned camera posterior as if it were a fresh measurement.

The direct raw-residual joint filter remains formally cleaner whenever it is computationally feasible. The pose-summary fallback remains usable for a first prototype or ablation study, but it should be documented as a weaker approximation whose neglected couplings and prior-reuse hazards can make the filter overconfident. This places the practical hierarchy in a clear order: exact joint Bayes first, message-based recursive approximation second, pose-summary shortcut last [8, 11–13, 25].

## 2.5 Synthesis and literature position

The reference problem is a posterior on a trajectory curve together with a static auxiliary-tag map in an anchored world frame. The prior is a law on that mixed object: smooth body motion, slowly drifting biases, and static but initially unknown tag poses. In the full joint model the IMU contributes inertial transition likelihoods and the camera contributes the raw-pixel likelihood through the tag projection model. If one solves everything jointly, the result is a direct Bayesian estimator over both trajectory and map. If one instead chooses a speed-friendly decomposition, the defensible reduced model is the event-driven message architecture: IMU packets are used exactly once in propagation, the camera module converts each frame into a prior-free visual message, and the joint recursive estimator assimilates that message while preserving pose–map cross-covariances for the active states. Pose pseudo-measurements remain only as a weaker fallback. The practical message is therefore simple: exact Bayes is the reference model, and recursive or fixed-lag estimators are computational approximations to it.

### 2.5.1 How the literature positions these choices

The Bayesian inverse-problem perspective treats trajectory and map as latent variables inferred from sensor data and priors [4]. Recursive Bayesian filtering and smoothing provide the general algorithmic backbone [7]. In visual–inertial robotics, filtering families such as MSCKF keep a Gaussian posterior over the current state and selected pose history [8]. Optimization families such as IMU-preintegrated MAP smoothing, OKVIS, and VINS-Mono instead solve nonlinear MAP problems over keyframes or sliding windows of recent states, sometimes with structureless visual factors rather than explicit landmark states [11, 12, 26, 27]. Fiducial-tag systems such as TagSLAM explicitly represent static tag poses and body poses together in a factor graph [25]. Continuous-time Bayesian approaches such as STEAM place priors directly on the space of trajectories and then condition on observations [14, 28]. The anchored multi-tag method developed in this report fits naturally in this landscape as either a joint fiducial visual–inertial SLAM estimator or a conservative event-driven recursive approximation with a camera-message layer.

# Appendix A

# Prior samples for interpretation and sanity checking

| New notation | Meaning |
|---|---|
| $\Delta t$ | Discrete time step used for sampling; here $\Delta t = \frac{1}{60}$ s. |
| $N$ | Number of discrete states; in the annex $N = 241$ corresponding to a 4 s horizon. |
| $a_k, \alpha_k$ | Sampled translational acceleration and angular-acceleration drives at step $k$. |
| $\omega_k$ | Latent body angular velocity used only to generate the pose process in this annex. |
| $\mathrm{rpy}_k$ | Roll–pitch–yaw angles extracted from $R_k$ only for visualization; they are *not* the coordinates used by the filter. |

This annex does not introduce a new inferential model. It samples a simple discrete-time approximation to the extended trajectory prior from Chapter 1. The purpose is pedagogical: it is much easier to understand the role of the prior hyperparameters once one sees typical paths they generate. In particular, the sampler keeps the latent angular velocity $\omega_k$ explicit so that the orientation process can be generated directly. This matches the extended-state prior discussion in Chapter 1; it is not literally the same object as the reduced IMU-conditioned filter recursion used later for online estimation.

For the simulation we use the discrete generative scheme

$$
\begin{aligned}
p_{k+1} &= p_k + v_k \Delta t + \tfrac{1}{2} a_k \Delta t^2, && \text{(A.1)}\\
v_{k+1} &= v_k + a_k \Delta t, && \text{(A.2)}\\
R_{k+1} &= R_k \operatorname{Exp}\big( [\omega_k]_\times \, \Delta t \big), && \text{(A.3)}\\
\omega_{k+1} &= \omega_k + \alpha_k \Delta t, && \text{(A.4)}\\
b_{g,k+1} &= b_{g,k} + \eta_k^{bg}, && \text{(A.5)}\\
b_{a,k+1} &= b_{a,k} + \eta_k^{ba}, && \text{(A.6)}
\end{aligned}
$$

with independent Gaussian drives

$$
\begin{aligned}
a_k &\sim \mathcal{N}(0, \sigma_a^2 I_3), \qquad \alpha_k \sim \mathcal{N}(0, \sigma_\alpha^2 I_3), && \text{(A.7)}\\
\eta_k^{bg} &\sim \mathcal{N}(0, \sigma_{bg,\mathrm{rw}}^2 I_3), \qquad \eta_k^{ba} \sim \mathcal{N}(0, \sigma_{ba,\mathrm{rw}}^2 I_3). && \text{(A.8)}
\end{aligned}
$$

The sampled state shown in the plots is the filter-style state

$$x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k}),$$

while the auxiliary variable $\omega_k$ is used internally only to generate the orientation process. For display we convert $R_k$ into roll, pitch, and yaw angles. This is a visualization choice only: the actual filter should still represent rotational uncertainty in local Lie coordinates or in a quaternion plus 3D error state.

## A.1 Prior values used for the illustrative samples

| **Quantity** | **Value** | **Role in the prior sampler** |
|---|---|---|
| Time step $\Delta t$ | 1/60 s | Matches the camera-frame discretization used throughout the report. |
| Horizon $T$ | 4 s | Short lab-style motion segment. |
| $\sigma_{p0}$ | $(0.01, 0.01, 0.01)$ m | Initial position spread. |
| $\sigma_{v0}$ | $(0.03, 0.03, 0.03)$ m/s | Initial velocity spread. |
| $\sigma_{\mathrm{rpy},0}$ | $(2, 2, 3)^\circ$ | Initial orientation spread, shown in Euler angles only for display. |
| $\sigma_{\omega 0}$ | $(4, 4, 5)^\circ$/s | Initial angular-velocity spread used to drive the pose process. |
| $\sigma_a$ | 0.10 m/s$^2$ | Translational acceleration roughness. |
| $\sigma_\alpha$ | $6^\circ$/s$^2$ | Angular-acceleration roughness. |
| $\sigma_{bg,0}$ | $(0.20, 0.20, 0.20)^\circ$/s | Initial gyroscope-bias spread. |
| $\sigma_{ba,0}$ | $(0.03, 0.03, 0.03)$ m/s$^2$ | Initial accelerometer-bias spread. |
| $\sigma_{bg,\mathrm{rw}}$ | $0.005^\circ$/s per step | Gyroscope-bias random-walk step scale. |
| $\sigma_{ba,\mathrm{rw}}$ | 0.001 m/s$^2$ per step | Accelerometer-bias random-walk step scale. |

**Table A.1:** Illustrative prior values used for the annex sampler. These values are not claimed to be universal; they are chosen to produce motions in a realistic phone-on-arm laboratory regime, with translation excursions on the order of a few tenths of a meter and orientation excursions on the order of a few tens of degrees over the 4 s horizon.

The script used to generate the figures in this annex is stored alongside the report as `sample_prior_paths.py`, together with the JSON file `prior_sample_parameters.json`. The figures can therefore be regenerated whenever the assumed prior regime is adjusted.

## A.2 Three-dimensional prior samples

Figure A.1 shows eight sampled translation paths. The curves begin near the origin because the initial position prior is tight, then fan out as the sampled accelerations accumulate. This picture is the translational part of the prior alone; it is *not* a posterior trajectory and it is not conditioned on data.

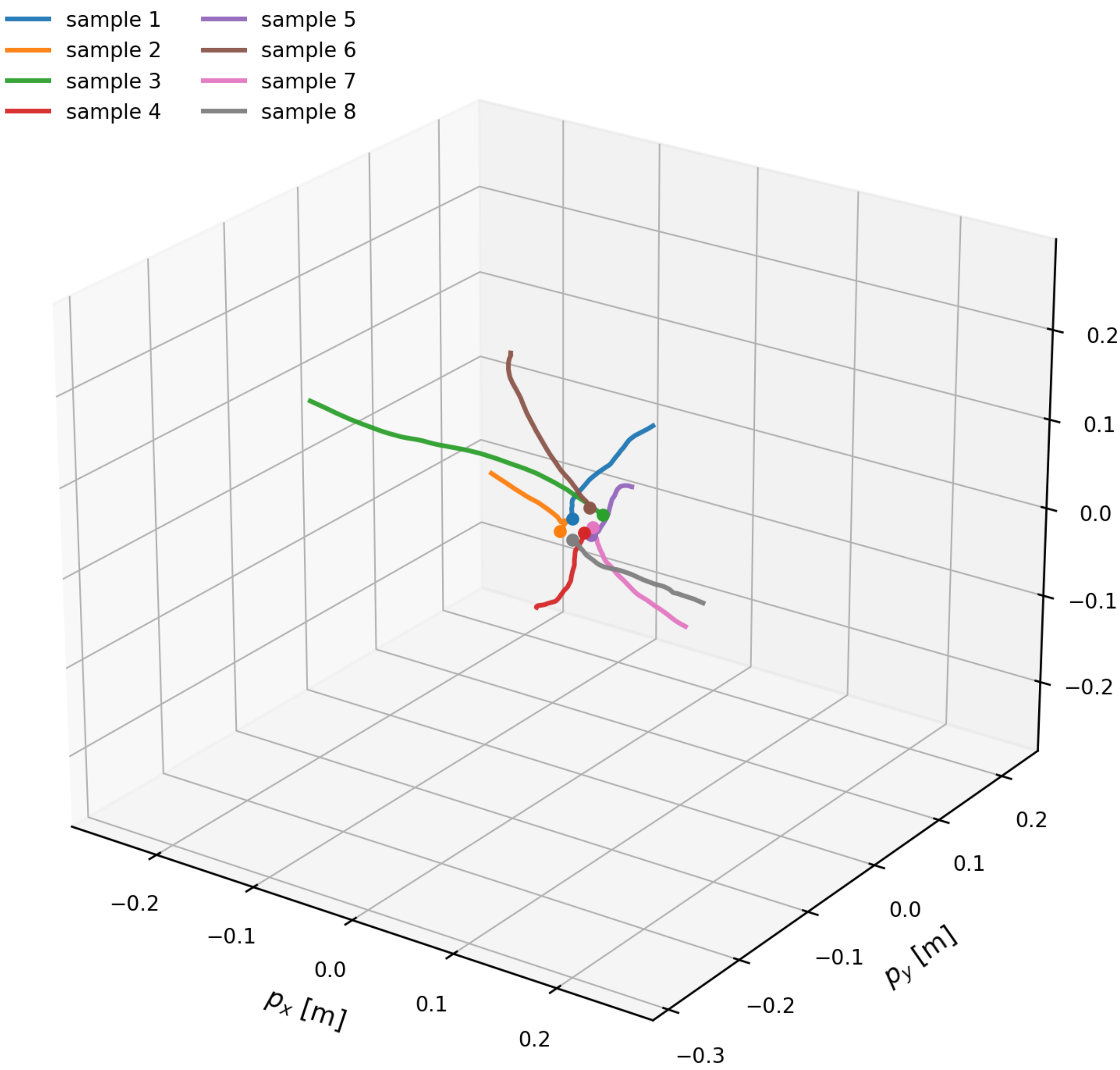


**Figure A.1:** Eight sample trajectories drawn from the discretized prior over a 4 s interval at 60 Hz. The paths illustrate the translational component of the prior before any measurement update is applied.

To visualize that the prior actually lives on poses rather than on positions only, Figure A.2 overlays body-frame triads along one sample path. The translational path lies in $\mathbb{R}^3$, but the orientation evolves on SO(3) and is injected through the exponential-map update.

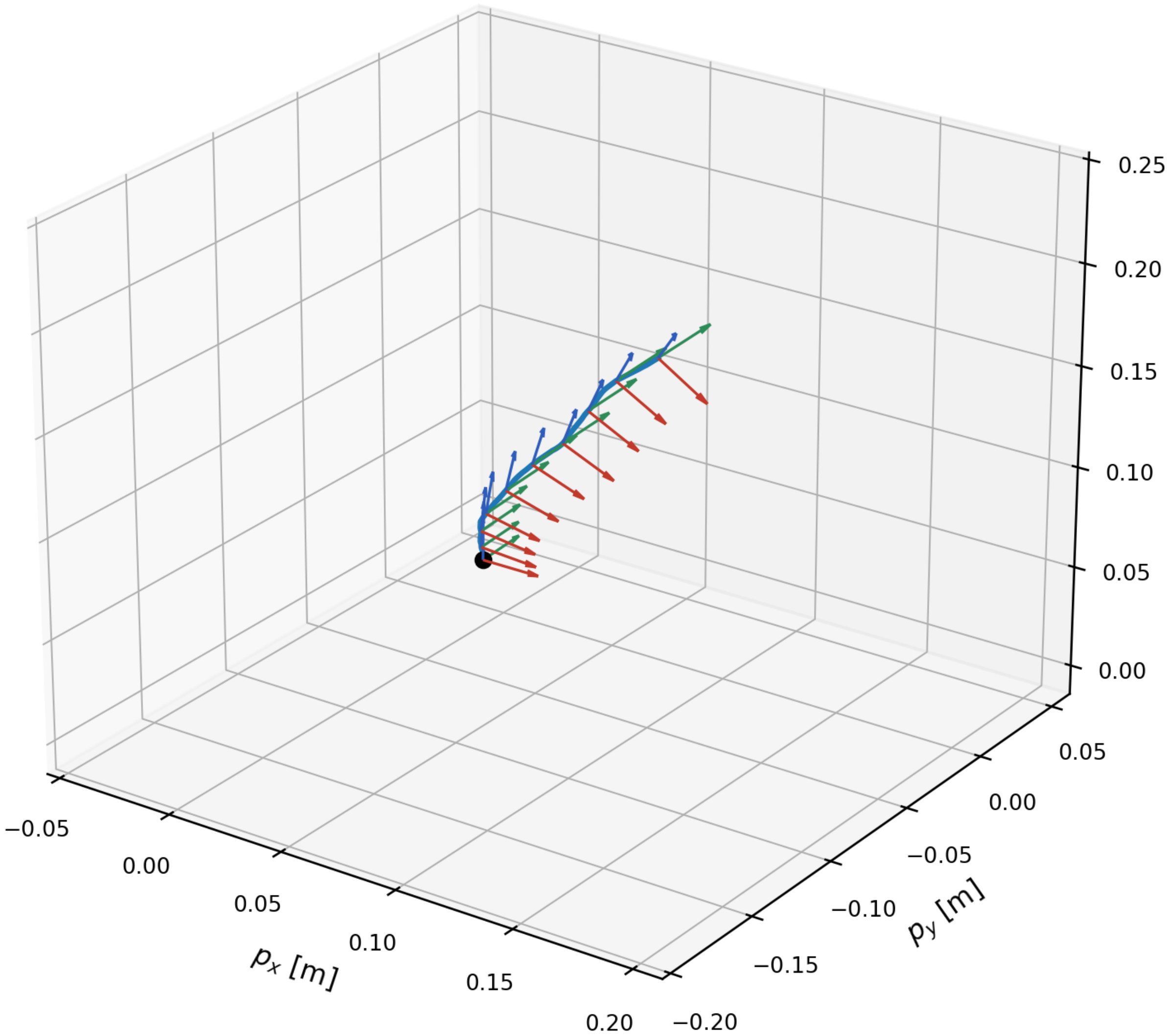


**Figure A.2:** One sampled pose trajectory with body-frame triads along the path. The translation is a path in $\mathbb{R}^3$, while the orientation component evolves on SO(3) and is updated multiplicatively.

## A.3 State evolution of all filter variables

Figure A.3 shows the evolution of all entries of the filter-style state

$$x_k = (R_k, p_k, v_k, b_{g,k}, b_{a,k}).$$

For readability, the orientation is displayed as roll, pitch, and yaw angles. The plots make the mixed geometry of the state visible: positions and velocities behave like ordinary Euclidean random walks with smoothness; the bias states evolve much more slowly; and the orientation display coordinates drift in a way that reflects accumulated angular velocity, even though the actual internal pose update is performed on the Lie group rather than by angle-wise addition.

## A.4 How these samples should be interpreted

The figures in this annex are useful because they connect the abstract prior hyperparameters to visible trajectory behavior.

- Increasing $\sigma_a$ makes the translational paths rougher and increases position spread rapidly over time.

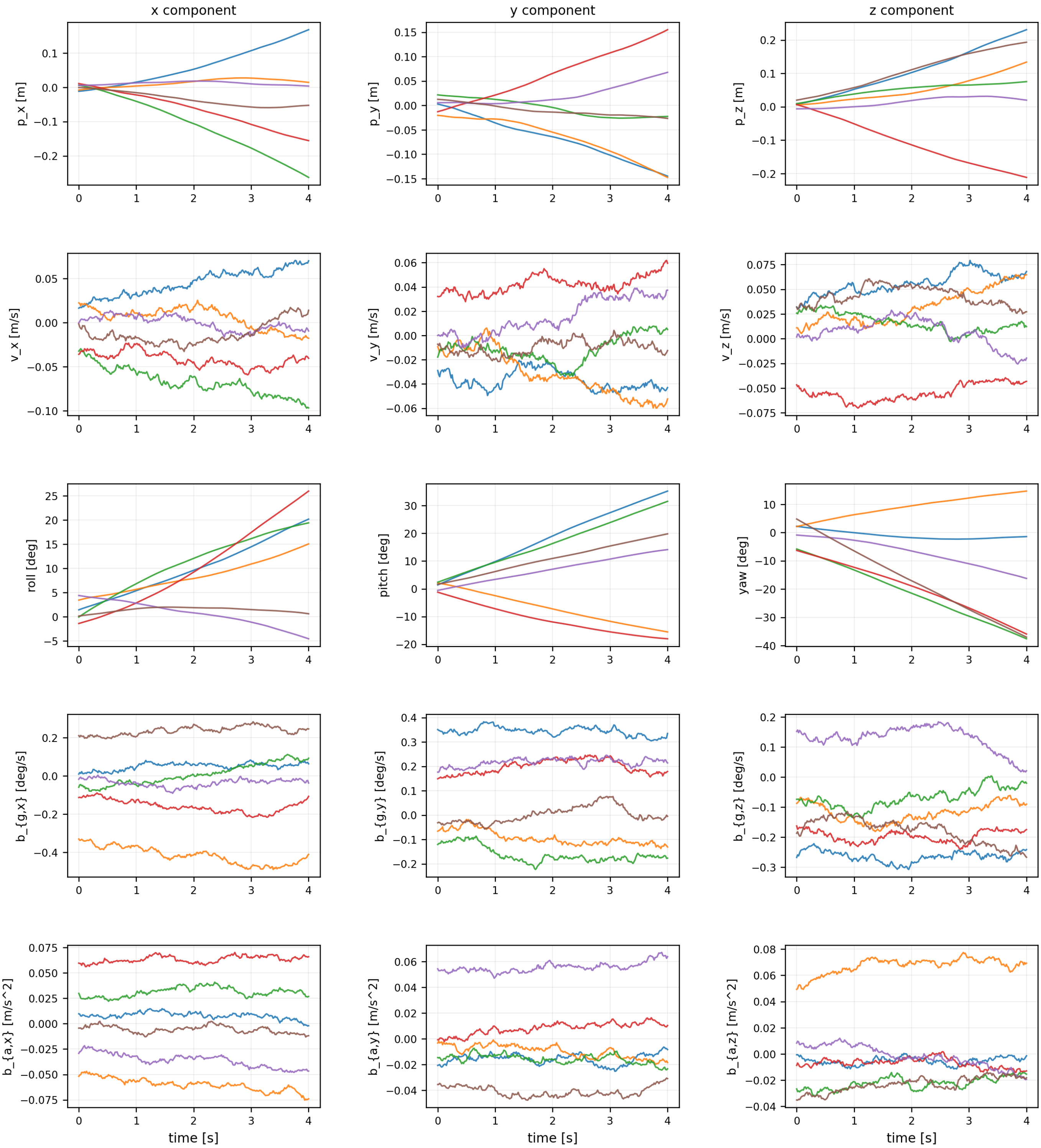


**Figure A.3:** Evolution of all filter-state entries for six prior samples. The third row displays orientation through roll, pitch, and yaw angles for visualization only. In the actual Bayesian formulation, the orientation posterior is represented locally in tangent coordinates or through a quaternion-plus-error representation rather than by filtering Euler angles directly.

- Increasing $\sigma_\alpha$ makes the orientation process wander more aggressively.
- Increasing $\sigma_{bg,\mathrm{rw}}$ or $\sigma_{ba,\mathrm{rw}}$ produces visibly drifting bias traces, which later appear in the filter as slower but persistent nuisance evolution.
- Tightening the initial spreads $\sigma_{p0}, \sigma_{v0}, \sigma_{\mathrm{rpy},0}$ shrinks the cloud near the start of the sequence, but it does not remove dynamic uncertainty later in time.

This is exactly why the prior deserves as much attention as the likelihood. Before any measurements are assimilated, the prior already says what kinds of motion are considered plausible. In a Kalman filter or IEKF, the same assumptions reappear as the initial covariance and the process-noise matrices. In a fixed-lag smoother or factor graph, they reappear as Gaussian motion factors. In a path-space Bayesian formulation, they are the law that generates the random curve itself.

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